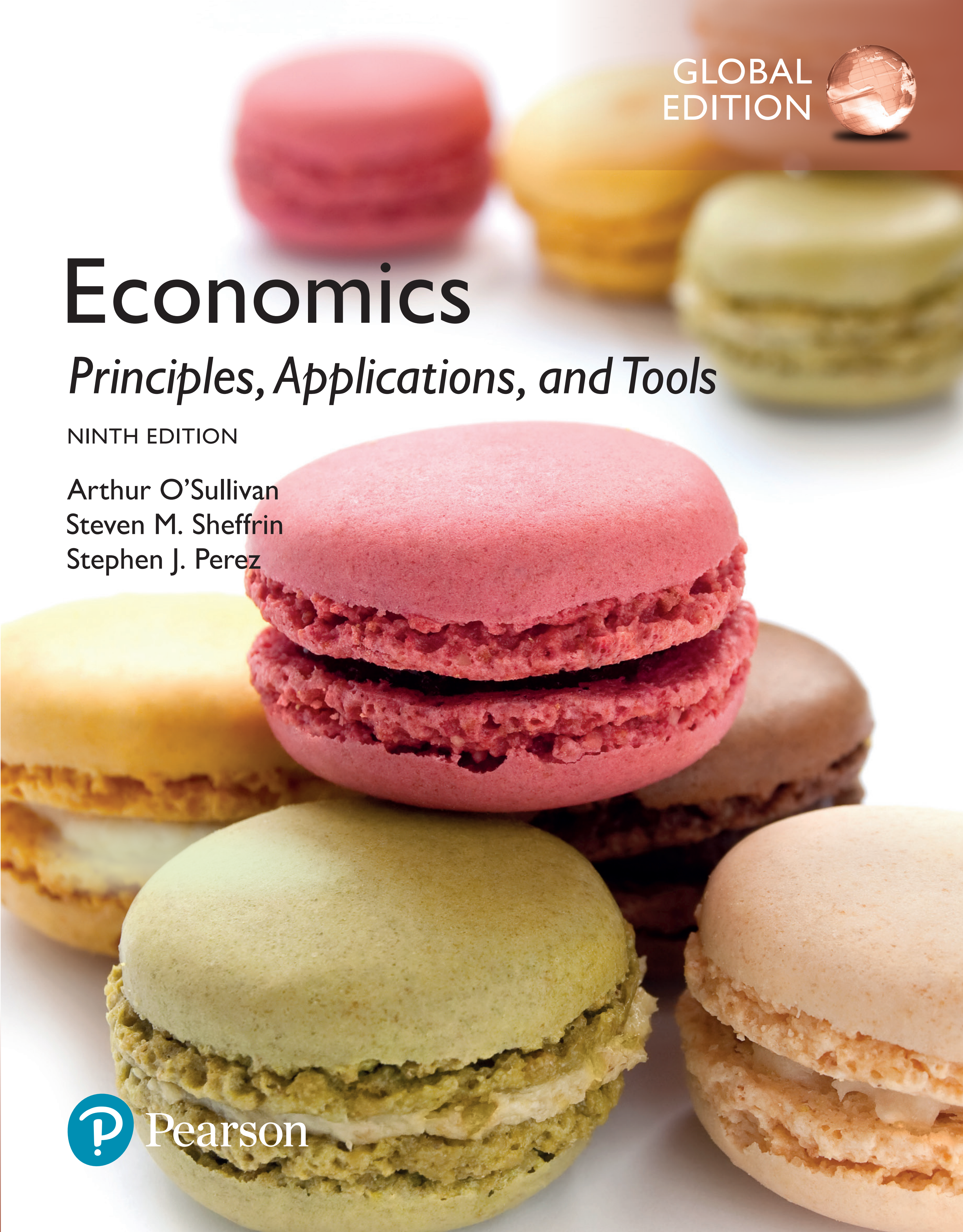
GLOBAL EDITION

Economics

Principles, Applications, and Tools

NINTH EDITION

Arthur O'Sullivan
Steven M. Sheffrin
Stephen J. Perez



Pearson

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Lewis and Clark College

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California State University, Sacramento

Pearson

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Authorized adaptation from the United States edition, entitled Economics: Principles, Applications, and Tools, 9th edition, ISBN 978-0-13-407884-7, by Arthur O'Sullivan, Steven M. Sheffrin, and Stephen J. Perez, published by Pearson Education © 2017.

ISBN 10: 1-292-16559-6
ISBN 13: 978-1-292-16559-2

British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library

10 9 8 7 6 5 4 3 2 1
14 13 12 11 10

About the Authors

ARTHUR O'SULLIVAN

is a professor of economics at Lewis and Clark College in Portland, Oregon. After receiving his B.S. in economics at the University of Oregon, he spent two years in the Peace Corps, working with city planners in the Philippines. He received his Ph.D. in economics from Princeton University in 1981 and has taught at the University of California, Davis, and Oregon State University, winning teaching awards at both schools. He is the author of the best-selling textbook *Urban Economics*, currently in its eighth edition, with translations into Russian, Chinese, Korean, and Greek.

Professor O'Sullivan's research explores economic issues concerning urban land use, environmental protection, and public policy. His articles have appeared in many economics journals, including the *Journal of Urban Economics*, *Journal of Environmental Economics and Management*, *National Tax Journal*, *Journal of Public Economics*, and *Journal of Law and Economics*.

Professor O'Sullivan lives with his family in Portland, Oregon. For recreation, he enjoys hiking, kiteboarding, and squash.

STEVEN M. SHEFFRIN

is professor of economics and executive director of the Murphy Institute at Tulane University. Prior to joining Tulane in 2010, he was a faculty member at the University of California, Davis, and served as department chairman of economics and dean of social sciences. He has been a visiting professor at Princeton University, Oxford University, London School of Economics, and Nanyang Technological University, and he has served as a financial economist with the Office of Tax Analysis of the United States Department of the Treasury. He received his B.A. from Wesleyan University and his Ph.D. in economics from the Massachusetts Institute of Technology.

Professor Sheffrin is the author of 10 other books and monographs and over 100 articles in the fields of macroeconomics, public finance, and international economics. His most recent books include *Rational Expectations* (second edition) and *Property Taxes and Tax Revolts: The Legacy of Proposition 13* (with Arthur O'Sullivan and Terri Sexton).

Professor Sheffrin has taught macroeconomics and public finance at all levels, from general introduction to principles classes (enrollments of 400) to graduate classes for doctoral students. He is the recipient of the Thomas Mayer Distinguished Teaching Award in economics.

He lives with his wife Anjali (also an economist) in New Orleans, Louisiana, and has two daughters who have studied economics. In addition to a passion for current affairs and travel, he plays a tough game of tennis.

STEPHEN J. PEREZ

is a professor of economics and NCAA faculty athletics representative at California State University, Sacramento. After receiving his B.A. in economics at the University of California, San Diego, he was awarded his Ph.D. in economics from the University of California, Davis, in 1994. He taught economics at Virginia Commonwealth University and Washington State University before coming to California State University, Sacramento, in 2001. He teaches macroeconomics at all levels as well as econometrics, sports economics, labor economics, and mathematics for economists.

Professor Perez's research explores most macroeconomic topics. In particular, he is interested in evaluating the ability of econometric techniques to discover the truth, issues of causality in macroeconomics, and sports economics. His articles have appeared in many economics journals, including the *Journal of Monetary Economics*; *Econometrics Journal*; *Economics Letters*; *Journal of Economic Methodology*; *Public Finance and Management*; *Journal of Economics and Business*; *Oxford Bulletin of Economics and Statistics*; *Journal of Money, Credit, and Banking; Applied Economics*; and *Journal of Macroeconomics*.

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Preface

In preparing this ninth edition, we had three primary goals. First, we wanted to incorporate the sweeping changes in the United States and world economies we have all witnessed in the last several years, and the difficulties that the world economics have continued to experience in recovering from the severe economic downturn. Second, we strived to update this edition to reflect the latest exciting developments in economic thinking and make these accessible to new students of economics. Finally, we wanted to stay true to the philosophy of the textbook—using basic concepts of economics to explain a wide variety of timely and interesting economic applications.

▶ WHAT'S NEW TO THIS EDITION

In addition to updating all the figures and data, we made a number of other key changes in this edition. They include the following:

- At the beginning of each chapter, we carefully refined our Learning Objectives to match the contents of the chapter closely. These give the students a preview of what they will learn in each section of the chapter, facilitating their learning.
- We discuss in Chapter 6 whether the recent major recession permanently affected labor force participation.
- We discuss in Chapter 8 the position of the pessimists who think that technological progress has slowed down.
- We also introduce in Chapter 8 the idea of controlled experiments in economic policy, as these experiments have been very influential in recent policy developments.
- We discuss in Chapter 13 the rationale and application of "stress tests" as a new tool for financial regulation.
- We introduce Janet Yellen, the new Chair of the Federal Reserve, in Chapter 14, and discuss her prior experience before she assumed her current role
- We revised and expanded our discussion of the euro in Chapter 19, reflecting the serious challenges now facing the European Monetary Union, particularly with the experience of Greece.
- We also incorporated a total of 44 exciting new Applications into this edition, including five in the common chapters (Chapters 1–4), 22 in macroeconomics, and 17 in microeconomics. In addition, we incorporated a total of 14 new chapter-opening stories, including one in the common chapters, seven in macroeconomics, and six in microeconomics. These fresh applications and chapter openers show the widespread relevance of economic analysis.
- In the chapters common to macroeconomics and microeconomics, the new applications include housing prices in Cuba (Chapter 1), property rights in urban slums (Chapter 3), and the effects of winds from the Sahara Desert on the price of chocolate (Chapter 4).
- In the macroeconomics chapters, the new applications include understanding the links between unemployment and unemployment insurance (Chapter 6), how the government promotes high levels of savings in Singapore (Chapter 7), how Greek citizens hoarded euros as the talk of crisis grew (Chapter 13), new research on " underwater" homeowners (Chapter 12), the debate on secular stagnation (Chapter 15), and how accounting for "missing" international financial flows changes our thinking of global investment patterns (Chapter 19).
- In the microeconomics chapters, the new applications include the market effects of a luxury tax (Chapter 21), the neuroscience of the "cola wars" between Coke and Pepsi (Chapter 22), the time path of blueberry prices triggered by publicity about the health benefits of eating blueberries (Chapter 24), the new advertising program for dairy products, "Milk Life" (Chapter 27), genetic discrimination in insurance (Chapter 29), clearing space debris (Chapter 30), responding to climate change by washing carbon out of the air (Chapter 31), and proposals to expand the earned income tax credit (Chapter 32).

▶ APPLYING THE CONCEPTS

This is an Applications-driven textbook. We carefully selected over 130 real-world Applications that help students develop and master essential economic concepts. Here is an example of our approach from Chapter 4, "Demand, Supply, and Market Equilibrium."

Application 1

THE LAW OF DEMAND FOR YOUNG SMOKERS

APPLYING THE CONCEPTS #1: What is the law of demand?

As price decreases and we move downward along the market demand for cigarettes, the quantity of cigarettes demanded increases for two reasons. First, people who smoked cigarettes at the original price respond to the lower price by smoking more. Second, some people start smoking.

In the United States, cigarette taxes vary across states, and studies of cigarette consumption patterns show that higher taxes mean less cigarette consumption by youths. Using data from the Youth Risk Behavior Surveys (YSBS), one study shows that increases in state cigarette taxes between 1990 and 2005 resulted in less participation (fewer smokers) and lower frequency (fewer cigarettes per smoker).

A change in cigarette taxes in Canada illustrates the second effect, the new-smoker effect. In 1994, several provinces in eastern Canada cut their cigarette taxes in response to the smuggling of cigarettes from the United States (where taxes are lower), and the price of cigarettes in the provinces decreased by roughly 50 percent. Researchers tracked the choices of 591 youths from the Waterloo Smoking Prevention Program and concluded that the lower price increased the smoking rate by roughly 17 percent. **Related to Exercises 1.6 and 1.8.**

SOURCES: (1) Anindya Sen and Tony Wirjanto, "Estimating the Impacts of Cigarette Taxes on Youth Smoking Participation, Initiation, and Persistence: Empirical Evidence from Canada," *Health Economics* 19 (2010), pp. 1264–1280. (2) Christopher Carpentera and Philip J. Cook, "Cigarette Taxes and Youth Smoking: New Evidence from National, State, and Local Youth Risk Behavior Surveys," *Journal of Health Economics* 27 (2008), pp. 287–299.

Each chapter includes three to five thought-provoking Applying the Concepts questions that convey important economic concepts, paired with and illustrated by an Application that discusses the concept and conveys its real-world use.

For each Application and Applying the Concepts question, we provide end-of-chapter exercises that test students' understanding of the concepts.

KEY TERMS

featherbedding, p. 715
income effect for leisure demand, p. 708
input-substitution effect, p. 706
labor union, p. 715
learning effect, p. 714
long-run demand curve for labor, p. 706
marginal product of labor, p. 704
marginal-revenue product of labor (*MRP*), p. 704
market supply curve for labor, p. 709
means-tested programs, p. 720
output effect, p. 706
short-run demand curve for labor, p. 704
signaling effect, p. 714
substitution effect for leisure demand, p. 708

EXERCISES All problems are assignable in MyLab Economics.

The Demand for Labor

Explain why competition generates wages equal to marginal revenue product.

1.1 The marginal revenue product of labor equals ________ times ________.
1.2 A profit-maximizing firm will hire the number of workers where ________ equals ________.
1.3 Your favorite professional team is considering hiring a new player for $3 million per year. It will be sensible (profitable) to hire the player if his ________ is greater than the $3 million cost.
1.4 Arrows up or down: The logic of the output effect is that a decrease in the wage will ________ production costs, so the price of output will ________ and the quantity of output demanded will ________. As a result, the quantity of labor demanded will ________.
1.5 The input-substitution effect is that a decrease in the wage ________ (increases/decreases) the quantity of labor per unit of ________, so the quantity of labor demanded ________ (increases/decreases).
1.6 The short-run market demand curve for labor is ________ (steeper/flatter) than the long-run demand because ________ occur(s) in the short run.
1.7 Fill the blanks with 75, 100, 117, 200, or 360. In major league baseball, the marginal revenue product of the typical free agent is roughly ________ percent of his salary, compared to ________ percent for a journeyman and ________ percent for an apprentice. (Related to Application 1 on page 707.)
1.8 **Demand for News Kids.** Consider the market for newspaper delivery kids in Kidsville. Each news kid receives a piece rate of $2 per subscriber per month and has a fixed territory that initially has 100 subscribers. The price elasticity of demand for subscriptions is 2.0. Suppose the new city council of Kidsville passes a law that establishes a minimum piece rate of $3 per subscriber per month. As a result, the publisher increases the monthly price of a subscription by 20 percent. How will the new law affect the monthly income of the typical news kid?
1.9 **Demand for Airline Pilots.** Comment on the following: "There is no substitute for an airline pilot: Someone has to fly the plane. Therefore, an increase in the wage of airline pilots will not change the number of pilots used by the airlines."

The Supply of Labor

Explain why an increase in the wage could increase, decrease, or not change hours worked.

2.1 Arrows up or down: An increase in the wage ________ the opportunity cost of leisure time, which tends to ________ leisure time and ________ labor time.
2.2 Arrows up or down: An increase in the wage ________ real income, and if leisure is a normal good this tends to ________ leisure time and ________ labor time.
2.3 We ________ (can/cannot) predict a worker's response to an increase in the wage because the ________ effect and the ________ effect work in ________ (the same/opposite) direction(s).
2.4 Your objective is to earn exactly $120 per week. If your wage decreases from $6 to $4 per hour, you respond by working ________ hours instead of ________ hours. In other words, your labor-supply curve is ________ sloped.
2.5 In the short run, labor supply is influenced by the wage rate, while in the long run, it is influenced by net advantages of the job. ________ (True/False)

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4.10 **Crop Insurance.** Consider a state in which farmers are divided equally into two types: high risk and low risk. The average annual crop loss (and possible insurance claim) is $200 for a low-risk farmer and $1,200 for a high-risk farmer.
a. If all farmers were to buy insurance, what is the break-even price for the insurance company?
b. Suppose a farmer will purchase insurance only if the price (the annual premium) is no more than 50 percent higher than his or her average crop loss. What is the equilibrium price?

4.11 **Safety Rebate from the Insurance Company.** In 2010 a leading insurance company started a policy that pays a policyholder a 5 percent rebate on his or her insurance premium in any year in which the driver does not file an insurance claim. For example, a household with an annual premium of $1,200 will get a $60 rebate check each year it does not file a claim. Explain the rationale for the rebate policy. What problem is the policy trying to solve? (Related to Application 4 on page 653.)

The Economics of Consumer Search

Use the marginal principle to describe optimal search by consumers.

5.1 The reservation price is the price at which the consumer is ________ about additional search, meaning that the ________ of search equals the ________.
5.2 If a consumer knows that TVs are available at BestBuy at a price as low as $200, it is sensible not to continue shopping until the consumer finds the lowest-price store. ________ (True/False)
5.3 Suppose the range of prices for a cooking book is $30 to $70 and your discovered price (lowest price so far) is $48. If you visit one more store and discover a lower price, the best guess for a lower price is ____ the savings if you discover this lower price is
5.4 As the opportunity cost of search time in reservation price ________ (increases, and the amount of search time decreases).
5.5 A doubling of prices ________ (increases the marginal benefit of search at a given price, and the percentage gap between th tion price and the lowest price ________ decreases).
5.6 A study of consumer search for liquid dete mated that a doubling of income decreased of search by roughly ________ (1, 14, 50, 7 (Related to Application 5 on page 657.)
5.7 **Compute the Reservation Price.** Suppos of prices for a consumer good is $20 to $ prices in this range are equally likely.
a. Fill in the blanks in the table.

Discovered price (lowest so far)	$40
Probability of discovering lower price in next visit	____
Best guess of lower price	____
Savings if lower price is discovered	____
Marginal benefit: Expected savings from additional visit	____

b. If the marginal cost of search is $1.25, t tion price is ________.
5.8 **Internet and Reservation Prices.** Consid try that initially has no Internet service or so consumers must travel to stores to get i about prices. Use a graph to show the effec ducing Internet service on the reservation consumer good.

ECONOMIC EXPERIMENT

ROLLING FOR LEMONS

In this experiment, students play the role of consumers purchasing used cars. Over half the used cars on the road (57%) are plums, and the remaining cars (43%) are lemons. Each consumer offers a price for a used car and then rolls a pair of dice to find out whether he or she gets a lemon or a plum. In general, rolling a big number is good news: To get a plum, you need to roll a big number. The higher the price you offer, the smaller the number you must roll to get a plum. Here is how the experiment works:

- Each consumer tells the instructor how much he or she is offering for a used car and then rolls the dice.
- The instructor tells the consumer whether the number rolled is large enough to get a plum. If the number is not large enough, the consumer gets a lemon.
- The consumers' scores equal the difference b maximum amount they are willing to pay for car they got ($1,200 for a plum and $400 for a lemon) and the price they actually paid. For example, if Otto offers $500 and gets a plum, his score is $700. If Carla offers $600 and gets a lemon, her score is −$200.
- The instructor announces the result of each transaction to the class.
- There are three to five buying periods. At the end of the last trading period, each consumer adds up his or her score.

MyLab Economics
For additional economic experiments, please visit *www.myeconlab.com*.

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In addition, some chapters contain an Economic Experiment section that gives students the opportunity to do their own economic analysis.

▶ WHY FIVE KEY PRINCIPLES?

In Chapter 2, "The Key Principles of Economics," we introduce the following five key principles and then apply them throughout the book:

1. **The Principle of Opportunity Cost.** The opportunity cost of something is what you sacrifice to get it.
2. **The Marginal Principle.** Increase the level of an activity as long as its marginal benefit exceeds its marginal cost. Choose the level at which the marginal benefit equals the marginal cost.
3. **The Principle of Voluntary Exchange.** A voluntary exchange between two people makes both people better off.
4. **The Principle of Diminishing Returns.** If we increase one input while holding the other inputs fixed, output will increase, but at a decreasing rate.
5. **The Real-Nominal Principle.** What matters to people is the real value of money or income—its purchasing power—not the face value of money or income.

This approach of repeating five key principles gives students the big picture—the framework of economic reasoning. We make the key concepts unforgettable by using them repeatedly, illustrating them with intriguing examples, and giving students many opportunities to practice what they've learned. Throughout the text, economic concepts are connected to the five key principles when the following callout is provided for each principle:

PRINCIPLE OF OPPORTUNITY COST
The opportunity cost of something is what you sacrifice to get it.

▶ HOW IS THE BOOK ORGANIZED?

Chapter 1, "Introduction: What Is Economics?" uses three current policy issues—traffic congestion, poverty in Africa, and Japan's prolonged recession—to explain the economic way of thinking. Chapter 2, "The Key Principles of Economics," introduces the five principles we return to throughout the book. Chapter 3, "Exchange and Markets," is devoted entirely to exchange and trade. We discuss the fundamental rationale for exchange and introduce some of the institutions modern societies developed to facilitate trade.

Students need to have a solid understanding of demand and supply to be successful in the course. Many students have difficulty understanding movement along a curve versus shifts of a curve. To address this difficulty, we developed an innovative way to organize topics in Chapter 4, "Demand, Supply, and Market Equilibrium." We examine the law of demand and changes in quantity demanded, the law of supply and changes in quantity supplied, and then the notion of market equilibrium. After students have a firm grasp of equilibrium concepts, we explore the effects of changes in demand and supply on equilibrium prices and quantities. You can present either macroeconomics or microeconomics chapters first, depending on your preference.

Summary of the Macroeconomics Chapters

Part 2, "The Basic Concepts of Macroeconomics" (Chapters 5 and 6), introduces students to the key concepts—GDP, inflation, unemployment—that are used throughout the text and in everyday economic discussion. The two chapters in this section provide the building blocks for the rest of the book. Part 3, "The Economy in the Long Run" (Chapters 7 and 8), analyzes how the economy operates at full employment and explores the causes and consequences of economic growth.

Next we turn to the short run. We begin the discussion of business cycles, economic fluctuations, and the role of government in Part 4, "Economic Fluctuations and Fiscal Policy" (Chapters 9 through 12). We devote an entire chapter to the structure of government spending and revenues and the role of fiscal policy. In Part 5, "Money, Banking, and Monetary Policy" (Chapters 13 and 14), we introduce the key elements of both monetary theory and policy into our economic models. Part 6, "Inflation, Unemployment, and Economic Policy" (Chapters 15 through 17), brings the important questions of the dynamics of inflation and unemployment into our analysis. Finally, the last two chapters in Part 7, "The International Economy" (Chapter 18 and 19), provide an in-depth analysis of both international trade and finance.

A Few Features of Our Macroeconomics Chapters

The following are a few features of our macroeconomics chapters:

- **Flexibility.** A key dilemma confronting economics professors has always been how much time to devote to long-run topics, such as growth and production, versus short-run topics, such as economic fluctuations and business cycles. Our book is designed to let professors choose. It works like this: To pursue a long-run approach, professors should initially concentrate on Chapters 1 through 4, followed by Chapters 5 through 8.
- To focus on economic fluctuations, start with Chapters 1 through 4, present Chapter 5, "Measuring a Nation's Production and Income," and Chapter 6, "Unemployment and Inflation," and then turn to Chapter 9, "Aggregate Demand and Aggregate Supply."
- Chapter 11, "The Income-Expenditure Model," is self-contained, so instructors can either skip it completely or cover it as a foundation for aggregate demand.
- **Long Run.** Throughout most of the 1990s, the U.S. economy performed very well—low inflation, low unemployment, and rapid economic growth. This robust performance led to economists' increasing interest in trying to understand the processes of economic growth. Our discussion of economic growth in Chapter 8, "Why Do Economies Grow?" addresses the fundamental question of how long-term living standards are determined and why some countries prosper while others do not. This is the essence of economic growth. As Nobel Laureate Robert E. Lucas, Jr., once wrote, "Once you start thinking about growth, it is hard to think of anything else."
- **Short Run.** The great economic expansion of the 1990s came to an end in 2001, as the economy started to contract. The recession beginning in 2007 was the worst downturn since World War II. Difficult economic times remind us that macroeconomics is also concerned with understanding the causes and consequences of economic fluctuations. Why do economies experience recessions and depressions, and what steps can policymakers take to stabilize the economy and ease the devastation people suffer from them? This has been a constant theme of macroeconomics throughout its entire history and is covered extensively in the text.
- **Policy.** Macroeconomics is a policy-oriented subject, and we treat economic policy in virtually every chapter. We discuss both important historical and more recent macroeconomic events in conjunction with the theory. In addition, we devote Chapter 17, "Macroeconomic Policy Debates," to three important policy topics that recur frequently in macroeconomic debates: the role of government deficits, whether the Federal Reserve should target inflation or other objectives, and whether income or consumption should be taxed.

Summary of the Microeconomics Chapters

A course in microeconomics starts with the first four chapters of the book, which provide a foundation for more detailed study of individual decision making and markets.

Part 8, "A Closer Look at Demand and Supply," (Chapters 20 through 22), provides a closer look at demand and supply, including elasticity, market efficiency, and consumer choice. Part 9, "Market Structures and Pricing" (Chapters 23 through 28), starts with a discussion of production and costs, setting the stage for an examination of alternative market structures, including the extremes of perfect competition and monopoly, as well as the middle ground of monopolistic competition and oligopoly. The last chapter in Part 9 discusses antitrust policy and deregulation. Part 10, "Externalities and Information" (Chapters 29 through 31), discusses the circumstances under which markets break down, including imperfect information, public goods, and environmental degradation.

Part 11, "The Labor Market and Income Distribution" (Chapter 32), explores the economic forces that determine wages, and also examines recent changes in the distribution of income and the effects of government programs on the income distribution.

▶ MyLab Economics

Digital Features Located in MyLab Economics

MyLab Economics is a unique online course management, testing, and tutorial resource. It is included with the eText version of the book or as a supplement to the print book. Students and instructors will find the following online resources to accompany the ninth edition:

- **Concept Checks:** Each section of each learning objective concludes with an online Concept Check that contains one or two multiple choice, true/false, or fill-in questions. These checks act as "speed bumps" that encourage students to stop and check their understanding of fundamental terms and concepts before moving on to the next section. The goal of this digital resource is to help students assess their progress on a section-by-section basis, so they can be better prepared for homework, quizzes, and exams.
- **Animations:** Graphs are the backbone of introductory economics, but many students struggle to understand and work with them. Many of the numbered figures in the text a supporting animated version online. The goal of this digital resource is to help students understand shifts in curves, movements along curves, and changes in equilibrium values. Having an animated version of a graph helps students who have difficulty interpreting the static version in the printed text. Graded practice exercises are included with the animations. Our experience is that many students benefit from this type of online learning.
- **Graphs Updated with Real-Time Data from FRED:** Approximately 16 graphs are continuously updated online with the latest available data from FRED (Federal Reserve Economic Data), which is a comprehensive, up-to-date data set maintained by the Federal Reserve Bank of St. Louis.

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  Students can display a pop-up graph that shows new data plotted in the graph. The goal of this digital feature is to help students understand how to work with data and understand how including new data affects graphs.
- **Interactive Problems and Exercises Updated with Real-Time Data from FRED:** The end-of-chapter problems in select chapters include real-time data exercises that use the latest data from FRED. The book contains several of these specially-selected exercises. The goal of this digital feature is to help students become familiar with this key data source, learn how to locate data, and develop skills in interpreting data.

▶ INTEGRATED SUPPLEMENTS

The authors and Pearson Education have worked together to integrate the text and media resources to make teaching and learning easier.

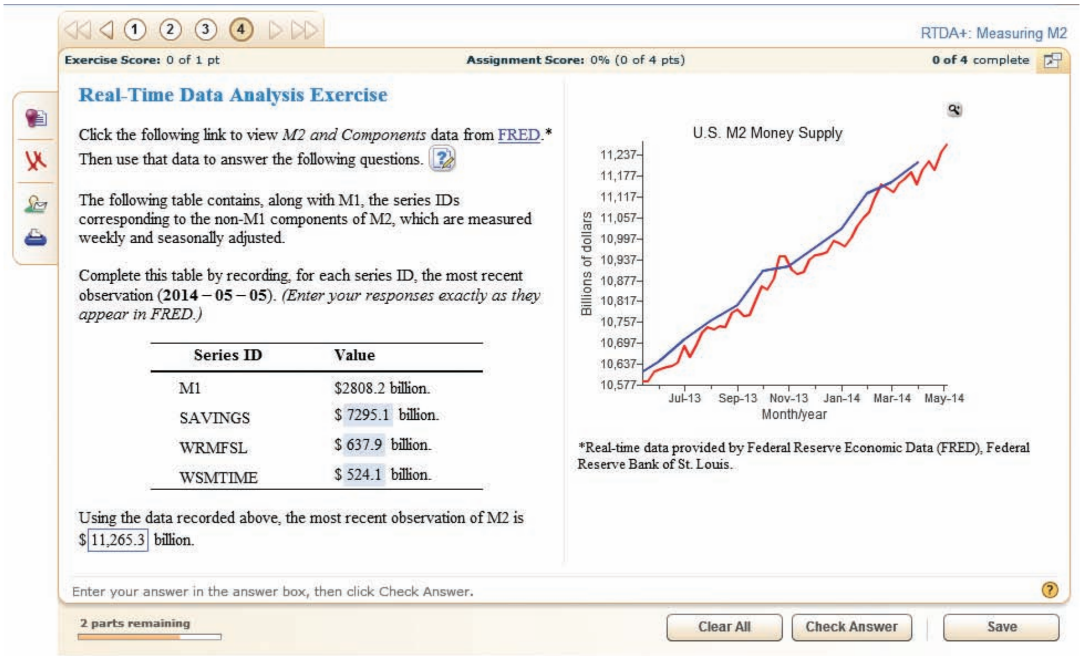
For the Instructor

Instructors can choose how much or how little time to spend setting up and using MyLab Economics.

Each chapter contains two preloaded exercise sets that can be used to build an individualized study plan for each student. These study plan exercises contain tutorial resources, including instant feedback, links to the appropriate learning objective in the eText, pop-up definitions from the text, and step-by-step guided solutions, where appropriate. After the initial setup of the course by the instructor, student use of these materials requires no further instructor setup. The online grade book records each student's performance and time spent on the tests and study plan and generates reports by student or chapter.

Instructors can fully customize MyLab Economics to match their course exactly, including reading assignments, homework assignments, video assignments, current news assignments, and quizzes and tests. Assignable resources include:

- Preloaded exercise assignments sets for each chapter that include the student tutorial resources mentioned earlier
- Preloaded quizzes for each chapter that are unique to the text and not repeated in the study plan or homework exercise sets
- Study plan problems that are similar to the end-of-chapter problems and numbered exactly like the book to make assigning homework easier
- Real-Time-Data Analysis Exercises, marked with , allow students and instructors to use the very latest data from FRED. By completing the exercises, students become familiar with a key data source, learn how to locate data, and develop skills in interpreting data.
- In the eText available in MyLab Economics, select figures labeled MyLab Economics Real-time data allow students to display a pop-up graph updated with real-time data from FRED.
- Current News Exercises provide a turnkey way to assign gradable news-based exercises in MyLab Economics. Each week, Pearson scours the news, finds a current microeconomics and macroeconomics article, creates exercises around these news articles, and then automatically adds them to MyLab Economics. Assigning and grading current news-based exercises that deal with the latest micro and macro events and policy issues has never been more convenient.
- Experiments in MyLab Economics are a fun and engaging way to promote active learning and mastery of important economic concepts. Pearson's Experiments program is flexible, easy-to-assign, auto-graded, and available in Single and Multiplayer versions.
- Single-player experiments allow your students to play against virtual players from anywhere at any time so long as they have an Internet connection.



- Multiplayer experiments allow you to assign and manage a real-time experiment with your class.
- Pre- and post-questions for each experiment are available for assignment in MyLab Economics.

For a complete list of available experiments, visit **http://www.myeconlab.com.**

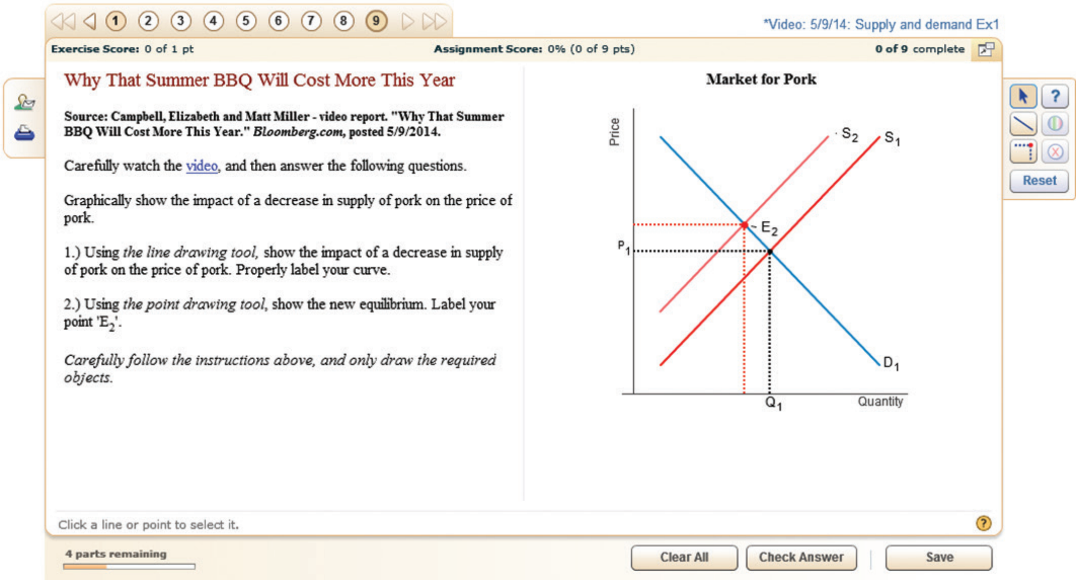
- Test Item File questions that allow you to assign quizzes or homework that will look just like your exams
- Econ Exercise Builder, which allows you to build customized exercises

Exercises include multiple-choice, graph drawing, and free-response items, many of which are generated algorithmically so that each time a student works them, a different variation is presented.

MyLab Economics grades every problem type except essays, even problems with graphs. When working homework exercises, students receive immediate feedback, with links to additional learning tools.

Customization and Communication

MyLab Economics in Pearson MyLab/Mastering provides additional optional customization and communication tools. Instructors who teach distance-learning courses or very large lecture sections find the Pearson MyLab/Mastering format useful because they can upload course documents and assignments, customize the order of chapters, and use



communication features such as Document Sharing, Chat, ClassLive, and Discussion Board.

For the Student

MyLab Economics puts students in control of their learning through a collection of testing, practice, and study tools tied to the online, interactive version of the textbook and other media resources.

Students can study on their own or can complete assignments created by their instructor. In MyLab Economics's structured environment, students practice what they learn, test their understanding, and pursue a personalized study plan generated from their performance on sample tests and from quizzes created by their instructors. In Homework or Study Plan mode, students have access to a wealth of tutorial features, including:

- Instant feedback on exercises that helps students understand and apply the concepts
- Links to the eText to promote reading of the text just when the student needs to revisit a concept or an explanation
- Step-by-step guided solutions that force students to break down a problem in much the same way an instructor would do during office hours
- Pop-up key term definitions from the eText to help students master the vocabulary of economics
- A graphing tool that is integrated into the various exercises to enable students to build and manipulate graphs to better understand how concepts, numbers, and graphs connect.

Additional MyLab Economics Tools

MyLab Economics includes the following additional features:

- **eText**—Students actively read and learn, and with more engagement than ever before, through embedded and auto-graded practice, real-time data-graph updates, animations, author videos, and more.
- **Glossary flashcards**—Every key term is available as a flashcard, allowing students to quiz themselves on vocabulary from one or more chapters at a time.

MyLab Economics content has been created through the efforts of Chris Annala, State University of New York–Geneseo; Charles Baum, Middle Tennessee State University; Peggy Dalton, Frostburg State University; Carol Dole, Jacksonville University; David Foti, Lone Star College; Sarah Ghosh, University of Scranton; Satyajit Ghosh, Universtity of Scranton; Melissa Honig, Pearson Education; Woo Jung, University of Colorado; Courtney Kamauf, Pearson Education; Chris Kauffman, University of Tennessee–Knoxville; Russell Kellogg, University of Colorado–Denver; Noel Lotz, Pearson Education; Katherine McCann, University of Delaware; Daniel Mizak, Frostburg State University; Christine Polek, University of Massachusetts–Boston; Mark Scanlan, Stephen F. Austin State University; Leonie L. Stone, State University of New York–Geneseo; and Bert G. Wheeler, Cedarville University.

▶ OTHER RESOURCES FOR THE INSTRUCTOR

Instructor's Manual

Jeff Phillips of Colby-Sawyer College revised the *Instructor's Manual* for the ninth edition. The *Instructor's Manual* is designed to help the instructor incorporate applicable elements of the supplement package. It contains the following resources for each chapter:

- Chapter Summary: a bulleted list of key topics in the chapter
- Learning Objectives
- Approaching the Material; student-friendly examples to introduce the chapter
- Chapter Outline: summary of definitions and concepts
- Teaching Tips on how to encourage class participation
- Summary and discussion points for the Applications in the main text
- New Applications and discussion questions
- Solutions to all end-of-chapter exercises.

The *Instructor's Manual* is available for download from the Instructor's Resource Center (accessible from **http://www.pearsonglobaleditions.com/osullivan**). The solutions to the end-of-chapter review questions and problems were prepared by the authors and Jeff Phillips.

Test Item Files

Jeff Phillips of Colby-Sawyer College prepared the *Test Item Files* for the ninth edition. The *Test Item File* includes approximately 6,000 multiple-choice, true/false, short-answer, and graphing questions. There are questions to support each key feature in the book. The *Test Item Files* are available for download from the Instructor's Resource Center (accessible from **http://www.pearsonglobaleditions.com/osullivan**). Test questions are annotated with the following information:

- **Difficulty:** 1 for straight recall, 2 for some analysis, 3 for complex analysis
- **Type:** multiple-choice, true/false, short-answer, essay
- **Topic:** the term or concept the question supports
- **Learning outcome**
- **AACSB** (see description that follows)
- **Page number** in the text.

The Association to Advance Collegiate Schools of Business (AACSB) The Test Item File author has connected select questions to the general knowledge and skill guidelines found in the AACSB Assurance of Learning Standards.

What Is the AACSB? AACSB is a not-for-profit corporation of educational institutions, corporations, and other organizations devoted to the promotion and improvement of higher education in business administration and accounting. A collegiate institution offering degrees in business administration or accounting may volunteer for AACSB accreditation review. The AACSB makes initial accreditation decisions and conducts periodic reviews to promote continuous quality improvement in management education. Pearson Education is a proud member of the AACSB and is pleased to provide advice to help you apply AACSB Assurance of Learning Standards.

What Are AACSB Assurance of Learning Standards? One of the criteria for AACSB accreditation is the quality of curricula. Although no specific courses are required, the AACSB expects a curriculum to include learning experiences in the following categories of Assurance of Learning Standards:

- Written and Oral Communication
- Ethical Understanding and Reasoning
- Analytical Thinking Skills
- Information Technology
- Diverse and Multicultural Work
- Reflective Thinking
- Application of Knowledge.

Questions that test skills relevant to these standards are tagged with the appropriate standard. For example, a question testing the moral questions associated with externalities would receive the Ethical Understanding and Reasoning tag.

How Can Instructors Use the AACSB Tags? Tagged questions help you measure whether students are grasping the course content that aligns with the AACSB guidelines noted earlier. This in turn may suggest enrichment activities or other educational experiences to help students achieve these skills.

TestGen

The computerized TestGen package allows instructors to customize, save, and generate classroom tests. The test program permits instructors to edit, add, or delete questions from the Test Item Files; analyze test results; and organize a database of tests and student results. This software allows for extensive flexibility and ease of use. It provides many options for organizing and displaying tests, along with search and sort features. The software and the Test Item Files can be downloaded from the Instructor's Resource Center (accessible from **http://www.pearsonglobaleditions.com/osullivan**).

PowerPoint Lecture Presentation

Two sets of PowerPoint slides, prepared by Brock Williams of Metropolitan Community College, are available:

1. A comprehensive set of PowerPoint slides can be used by instructors for class presentations or by students for lecture preview or review. These slides include all the graphs, tables, and equations in the textbook. Two versions are available—step-by-step mode, in which you can build graphs as you would on a blackboard, and automated mode, in which you use a single click per slide. Instructors can download these PowerPoint presentations from the Instructor's Resource Center (accessible from **http://www.pearsonglobaleditions.com/osullivan**).
2. A student version of the PowerPoint slides is available as .pdf files. This version allows students to print the slides and bring them to class for note taking. Instructors can download these PowerPoint presentations from the Instructor's Resource Center (accessible from **http://www.pearsonglobaleditions.com/osullivan**).

Learning Catalytics™

Learning Catalytics is a "bring your own device" Web-based student engagement, assessment, and classroom intelligence system. This system generates classroom discussion, guides lectures, and promotes peer-to-peer learning with real-time analytics. Students can use any device to interact in the classroom, engage with content, and even draw and share graphs.

To learn more, ask your local Pearson representative or visit **https://www.learningcatalytics.com**.

Digital Interactives

Focused on a single core topic and organized in progressive levels, each interactive immerses students in an assignable and auto-graded activity. Digital Interactives are also engaging lecture tools for traditional, online, and hybrid courses, many incorporating real-time data, data displays, and analysis tools for rich classroom discussions.

▶ OTHER RESOURCES FOR THE STUDENT

In addition to MyLab Economics, Pearson provides the following resources.

Dynamic Study Modules

With a focus on key topics, these modules work by continuously assessing student performance and activity in real time and, using data and analytics, provide personalized content to reinforce concepts that target each student's particular strengths and weaknesses.

PowerPoint Slides

For student use as a study aid or note-taking guide, PowerPoint slides, prepared by Brock Williams of Metropolitan Community College, can be downloaded from MyLab Economics or the Instructor's Resource Center (accessible from **http://www.pearsonglobaleditions.com/osullivan**) and made available to students. The slides include:

- All graphs, tables, and equations in the text
- Figures in step-by-step mode and automated modes, using a single click per graph curve
- End-of-chapter key terms with hyperlinks to relevant slides

▶ REVIEWERS OF PREVIOUS EDITIONS

A long road exists between the initial vision of an innovative principles text and the final product. Along our journey we participated in a structured process to reach our goal. We wish to acknowledge the assistance of the many people who participated in this process.

Alabama

Jim Payne, Calhoun Community College
James Swofford, University of South Alabama

Alaska

Paul Johnson, University of Alaska, Anchorage

Arizona

Basil Al-Hashimi, Mesa Community College, Red Mountain
Pete Mavrokordatos, Tarrant County College/University of Phoenix
Evan Tanner, Thunderbird, The American Graduate School of International Management
Donald Wells, University of Arizona

California

Antonio Avalos, California State University, Fresno
Collette Barr, Santa Barbara Community College
T. J. Bettner, Orange Coast College
Peter Boelman-Lopez, Riverside Community College
Matthew Brown, Santa Clara University
Jim Cobb, Orange Coast College
John Constantine, Sacramento City College
Peggy Crane, San Diego State University
Albert B. Culver, California State University, Chico
Jose L. Esteban, Palomar College
Gilbert Fernandez, Santa Rosa Junior College
E. B. Gendel, Woodbury University
Charles W. Haase, San Francisco State University
John Henry, California State University, Sacramento
George Jensen, California State University, Los Angeles
Janis Kea, West Valley College
Rose Kilburn, Modesto Junior College
Philip King, San Francisco State University
Anthony Lima, California State University, Hayward
Bret Mcmurran, Chaffey College
Jon J. Nadenichek, California State University, Northridge
Alex Obiya, San Diego City College
Jack W. Osman, San Francisco State University
Jay Patyk, Foothill College
Stephen Perez, California State University, Sacramento
Ratha Ramoo, Diablo Valley College
Greg Rose, Sacramento City College
Kurt Schwabe, University of California, Riverside
Terri Sexton, California State University, Sacramento
David Simon, Santa Rosa Junior College
Xiaochuan Song, San Diego Mesa College
Ed Sorensen, San Francisco State University
Susan Spencer, Santa Rosa Junior College
Linda Stoh, Sacramento City College
Rodney Swanson, University of California, Los Angeles
Daniel Villegas, California Polytechnic State University

Colombia

Michael Jetter, Universidad EIFIT

Colorado

Steve Call, Metropolitan State College of Denver

Connecticut

John A. Jascot, Capital Community Technical College
Stephen Rubb, Sacred Heart University

Delaware

Lawrence Stelmach, Delaware Valley College

Florida

Irma de Alonso, Florida International University
Jay Bhattacharya, Okaloosa-Walton Community College
Edward Bierhanzl, Florida A&M University
Eric P. Chiang, Florida Atlantic University
Martine Duchatelet, Barry University
George Greenlee, St. Petersburg College, Clearwater
Martin Markovich, Florida A&M University
Thomas McCaleb, Florida State University
Barbara Moore, University of Central Florida
Stephen Morrell, Barry University
Carl Schmertmann, Florida State University
Garvin Smith, Daytona Beach Community College
Noel Smith, Palm Beach Community College
Michael Vierk, Florida International University
Joseph Ward, Broward Community College, Central
Virginia York, Gulf Coast Community College
Andrea Zanter, Hillsborough Community College

Georgia

Scott Beaulier, Mercer College
Ashley Harmon, Southeastern Technical College
Steven F. Koch, Georgia Southern University
L. Wayne Plumly, Jr., Valdosta State University
Greg Trandel, University of Georgia

Hawaii

Barbara Ross-Pfeiffer, Kapiolani Community College

Idaho

Charles Scott Benson, Jr., Idaho State University
Tesa Stegner, Idaho State University

Illinois

Diane Anstine, North Central College
Rosa Lea Danielson, College of DuPage
Sel Dibooglu, Southern Illinois University
Linda Ghent, Eastern Illinois University
Gary Langer, Roosevelt University
Nampeang Pingkarawat, Chicago State University
Dennis Shannon, Belleville Area College
Chuck Sicotte, Rock Valley College

Indiana

John L. Conant, Indiana State University
Mousumi Duttaray, Indiana State University
Robert B. Harris, Indiana Univ. Purdue Univ. Indianapolis
James T. Kyle, Indiana State University
Virginia Shingleton, Valparaiso University

Iowa

Dale Borman, Kirkwood Community College
Jonathan O. Ikoba, Scott Community College
Saul Mekies, Kirkwood Community College, Iowa City

Kansas

Carl Parker, Fort Hays State University
James Ragan, Kansas State University
Tracy M. Turner, Kansas State University

Kentucky

David Eaton, Murray State University
John Robertson, University of Kentucky

Louisiana

John Payne Bigelow, Louisiana State University
Sang Lee, Southeastern Louisiana University
Richard Stahl, Louisiana State University

Maine

George Schatz, Maine Maritime Academy

Maryland

Carey Borkoski, Anne Arundel Community College
Gretchen Mester, Anne Arundel Community College
Irvin Weintraub, Towson State University

Massachusetts

Hans Despain, Nichols College
Brian Deuriarte, Middlesex Community College
Dan Georgianna, University of Massachusetts, Dartmouth
James E. Hartley, Mount Holyoke College
Marlene Kim, University of Massachusetts, Boston
Mark Siegler, Williams College
Gilbert Wolfe, Middlesex Community College

Michigan

Christine Amsler, Michigan State University
Bharati Basu, Central Michigan University
Norman Cure, Macomb Community College
Susan Linz, Michigan State University
Scanlon Romer, Delta College
Robert Tansky, St. Clair County Community College
Wendy Wysocki, Monroe Community College

Minnesota

Ihsuan Li, Minnesota State University, Mankato
Mike Mcilhon, Augsburg College
Richard Milani, Hibbing Community College

Mississippi

Billy L. Carson II, Itawamba Community College
Arlena Sullivan, Jones County Junior College

Missouri

Duane Eberhardt, Missouri Southern State College
David Gillette, Truman State University
Brad Hoppes, Southwest Missouri State University
Denise Kummer, St. Louis Community College
Steven M. Schamber, St. Louis Community College, Meramec
Elias Shukralla, St. Louis Community College, Meramec
Keith Ulrich, Valencia Community College
George Wasson, St. Louis Community College, Meramec

Nebraska

Debbie Gaspard, Southeast Community College
Theodore Larsen, University of Nebraska, Kearney
Timothy R. Mittan, Southeast Community College
Stanley J. Peters, Southeast Community College
Brock Williams, Metropolitan Community College

Nevada

Stephen Miller, University of Nevada, Las Vegas
Charles Okeke, College of Southern Nevada

New Jersey

Len Anyanwu, Union County College
Richard Comerford, Bergen Community College
John Graham, Rutgers University
Paul C. Harris, Jr., Camden County College
Calvin Hoy, County College of Morris
Taghi Ramin, William Paterson University
Brian de Uriarte, Middlesex County College

New Hampshire

Jeff Phillips, Colby-Sawyer College

New Mexico

Carl Enomoto, New Mexico State University

New York

Farhad Ameen, State University of New York, Westchester County Community College
Karijit K. Arora, Le Moyne College
Alex Azarchs, Pace University
Kathleen K. Bromley, Monroe Community College
Barbara Connelly, Westchester Community College
George Frost, Suffolk County Community College
Susan Glanz, St. John's University
Serge S. Grushchin, ASA College of Advanced Technology
Robert Herman, Nassau Community College
Christopher Inya, Monroe Community College
Marie Kratochvil, Nassau Community College
Marianne Lowery, Erie Community College
Jeannette Mitchell, Rochester Institute of Technology
Ted Muzio, St. John's University

Gray Orphee, Rockland County Community College
Craig Rogers, Canisius College
Fred Tyler, Fordham University
Ezgi Uzel, SUNY-Maritime
Michael Vardanyan, Binghamton University

North Carolina

Katie Canty, Cape Fear Community College
Lee Craig, North Carolina State University
Hossein Gholami, Fayetteville Technical Community College
Michael G. Goode, Central Piedmont Community College
Charles M. Oldham, Jr., Fayetteville Technical Community College
Randall Parker, East Carolina University
Julianne Treme, University of North Carolina, Wilmington
Diane Tyndall, Craven Community College
Chester Waters, Durham Technical Community College
James Wheeler, North Carolina State University

North Dakota

Scott Bloom, North Dakota State University

Ohio

Fatma Abdel-Raouf, Cleveland State University
Jeff Ankrom, Wittenberg University
Erwin Ehrardt, University of Cincinnati
Kenneth C. Fah, Ohio Dominican University
Taghi T. Kermani, Youngstown State University
Dandan Liu, Kent State University

Oklahoma

Jeff Holt, Tulsa Community College
Marty Ludlum, Oklahoma City Community College
Dan Rickman, Oklahoma State University

Oregon

Tom Carroll, Central Oregon Community College
Jim Eden, Portland Community College
John Farrell, Oregon State University
David Figlio, University of Oregon
Randy R. Grant, Linfield College
Larry Singell, University of Oregon

Pennsylvania

Kevin A. Baird, Montgomery County Community College
Charles Beem, Bucks County Community College
Ed Coulson, Pennsylvania State University
Tahany Naggar, West Chester University
Abdulwahab Sraiheen, Kutztown University

South Carolina

Donald Balch, University of South Carolina
Calvin Blackwell, College of Charleston
Janice Boucher Breuer, University of South Carolina
Bill Clifford, Trident Technical College
Frank Garland, Tri-County Technical College
Charlotte Denise Hixson, Midlands Technical College
Woodrow W. Hughes, Jr., Converse College
Miren Ivankovic, Southern Wesleyan University
Chirinjev Peterson, Greenville Technical College
Gary Stone, Winthrop University
Denise Turnage, Midlands Technical College
Chad Turner, Clemson University

South Dakota

Joseph Santos, South Dakota State University

Tennessee

Cindy Alexander, Pellissippi State University
Nirmalendu Debnath, Lane College
Quenton Pulliam, Nashville State Technical College
Rose Rubin, University of Memphis
Thurston Schrader, Southwest Tennessee Community College

Texas

Rashid Al-Hmoud, Texas Technical University
Mahamudu Bawumia, Baylor University
Steven Beckham, Amarillo College
Omar Belazi, Midland College
Jack Bucco, Austin Community College
Cindy Cannon, North Harris College
David L. Coberly, Southwest Texas State University
Ed Cohn, Del Mar College
Dean Drainey, St. Phillips College
Michael I. Duke, Blinn College
Ghazi Duwaji, University of Texas, Arlington
Harry Ellis, University of North Texas
S. Aun Hassan, Texas Tech University
Thomas Jeitschko, Texas A&M University
Delores Linton, Tarrant County Community College, Northwest
Jessica McCraw, University of Texas, Arlington
Randy Methenitis, Richland College
William Neilson, Texas A&M University
Michael Nelson, Texas A&M University
Rhey Nolan, Tyler Junior College
Paul Okello, University of Texas, Arlington
Joshua Pickrell, South Plains College
John Pisciotta, Baylor University
John Rykowski, Kalamazoo Valley Community College
Dave Shorrow, Richland College
Steve Schwiff, Texas A&M University, Commerce
James R. Vanbeek, Blinn College
Inske Zandvliet, Brookhaven College

Utah

Reed Gooch, Utah Valley University
Ali Hekmat, College of Eastern Utah
Glenn Lowell, Utah Valley University

Virginia

James Brumbaugh, Lord Fairfax Community College, Middleton Campus
Bruce Brunton, James Madison University
Michael G. Heslop, North Virginia Community College
George Hoffer, Virginia Commonwealth University
Melanie Marks, Longwood College
Thomas J. Meeks, Virginia State University
John Min, Northern Virginia Community College, Alexandria
Shannon K. Mitchell, Virginia Commonwealth University
Bill Reese, Tidewater Community College, Virginia Beach

Washington

William Hallagan, Washington State University
Charles S. Wassell, Jr., Central Washington University
Mark Wylie, Spokane Falls Community College

Australia

Hak Youn Kim, Monash University

▶ CLASS TESTERS

A special acknowledgment goes to the instructors who were willing to class-test drafts of early editions in different stages of development. They provided us with instant feedback on parts that worked and parts that needed changes:

Sheryl Ball, Virginia Polytechnic Institute and State University
John Constantine, University of California, Davis
John Farrell, Oregon State University
James Hartley, Mt. Holyoke College
Kailash Khandke, Furman College
Peter Lindert, University of California, Davis
Louis Makowski, University of California, Davis
Barbara Ross-Pfeiffer, Kapiolani Community College

▶ FOCUS GROUPS

We want to thank the participants who took part in the focus groups for the first and second editions; they helped us see the manuscript from a fresh perspective:

Carlos Aquilar, El Paso Community College
Jim Bradley, University of South Carolina
Thomas Collum, Northeastern Illinois University
David Craig, Westark College
Jeff Holt, Tulsa Junior College
Thomas Jeitschko, Texas A&M University
Gary Langer, Roosevelt University
Mark McLeod, Virginia Polytechnic Institute and State University
Tom McKinnon, University of Arkansas
Amy Meyers, Parkland Community College
Hassan Mohammadi, Illinois State University
John Morgan, College of Charleston
Norm Paul, San Jacinto Community College
Nampeang Pingkaratwat, Chicago State University
Scanlan Romer, Delta Community College
Barbara Ross-Pfeiffer, Kapiolani Community College
Zahra Saderion, Houston Community College
Virginia Shingleton, Valparaiso University
Jim Swofford, University of South Alabama
Janet West, University of Nebraska, Omaha
Linda Wilson, University of Texas, Arlington
Michael Youngblood, Rock Valley Community College

▶ GLOBAL EDITION CONTRIBUTORS

Wiktor Bolkunow, Warsaw School of Economics
Malgorzata Cieciora, Polish-Japanese Institute of Information Technology
Kwan Wai Ko, The Chinese University of Hong Kong
Mikael Lindback, Umeå University
Marina Binti Mustapha, Taylor's University
Gabriela Sterian, Romanian-American University

▶ GLOBAL EDITION REVIEWERS

Valentin Cojanu, Academy of Economics Studies in Bucharest
Yasodhara Das, The Foundation of Agrarian Studies
Rawindaran Nair, Cardiff University

▶ A WORLD OF THANKS . . .

We would also like to acknowledge the team of dedicated authors who contributed to the various ancillaries that accompany this book: Jeff Phillips of Colby-Sawyer College; Brian Rosario of California State University, Sacramento; and Brock Williams of Metropolitan Community College.

For the eighth edition, Liz Napolitano was the senior production project manager who worked with Michelle Gardner at SPi-Global to turn our manuscript pages into a beautiful published book. Lindsey Sloan, program manager, guided the project and coordinated the schedules for the book and the extensive supplement package that accompanies the book. David Alexander, executive acquisitions editor, supported us and our users during the life of this edition.

From the start, Pearson provided us with first-class support and advice. Over the first seven editions, many people contributed to the project, including Leah Jewell, Rod Banister, P. J. Boardman, Marie McHale, Gladys Soto, Lisa Amato, Victoria Anderson, Cynthia Regan, Kathleen McLellan, Sharon Koch, David Theisen, Steve Deitmer, Christopher Bath, Ben Paris, Elisa Adams, Jodi Bolognese, David Alexander, Virginia Guariglia, and Lynne Breitfeller.

Last but not least, we must thank our families, who have seen us disappear, sometimes physically and other times mentally, to spend hours wrapped up in our own world of principles of economics. A project of this magnitude is very absorbing, and our families have been particularly supportive in this endeavor.

Arthur O'Sullivan

Steven Sheffrin

Stephen Perez

CHAPTER

1 Introduction: What Is Economics?

Economics is the science of choice, exploring the choices made by individuals and organizations.

Over the last few centuries, these choices have led to substantial gains in the standard of living around the globe. In the United States, the typical person today has roughly seven times the income and purchasing power of a person 100 years ago. Our prosperity is the result of choices made by all sorts of people, including inventors, workers, entrepreneurs, and the people who saved money and loaned it to others to invest in machines and other tools of production. One reason we have prospered is greater efficiency: We have discovered better ways to use our resources—raw materials, time, and energy—to produce the goods and services we value.

As an illustration of changes in the standard of living and our growing prosperity, let's compare the way people listened to music in 1891 with how we listen today. You can buy an iPod shuffle® for $49 and fill it with 500 songs at $0.99 each. If you earn a wage of $15 per hour, it would take you about 36 hours of work to purchase and then fill an iPod. Back in 1891, the latest technological marvel was Thomas Edison's cylinder phonograph, which played music recorded on 4-inch cylinders. Imagine that you lived back then and wanted to get just as much music as you could fit on an iPod. Given the wages and prices in 1891, it would take you roughly 800 hours of work to earn enough money to buy the phonograph and all the cylinders. And if you wanted to keep your music with you, you would need 14 backpacks to carry the cylinders.

Although prosperity and efficiency are widespread, they are not universal. In some parts of the world, many people live in poverty. For example, in sub-Saharan Africa 388 million people—about half the population—live on less than $1.25 per day. And in all nations of the world, inefficiencies still exist, with valuable resources being wasted. For example, each year the typical urban commuter in the United States wastes more than 47 hours and $84 worth of gasoline stuck in rush-hour traffic.

CHAPTER OUTLINE AND LEARNING OBJECTIVES

MyLab Economics
MyLab Economics helps you master each objective and study more efficiently.

Economics provides a framework to diagnose all sorts of problems faced by society and then helps create and evaluate various proposals to solve them. Economics can help us develop strategies to replace poverty with prosperity, and to replace waste with efficiency. In this chapter, we explain what economics is and how we all can use economic analysis to think about practical problems and solutions.

What Is Economics?

Learning Objective 1.1

List the three key economic questions.

Economists use the word ***scarcity*** to convey the idea that resources—the things we use to produce goods and services—are limited, while human wants are unlimited. Therefore, we cannot produce everything that everyone wants. As the old saying goes, you can't always get what you want. **Economics** studies the choices we make when there is scarcity; it is all about trade-offs. Here are some examples of scarcity and the trade-offs associated with making choices:

scarcity
The resources we use to produce goods and services are limited.

economics
The study of choices when there is scarcity.

- You have a limited amount of time. If you take a part-time job, each hour on the job means one fewer hour for study or play.
- A city has a limited amount of land. If the city uses an acre of land for a park, it has one fewer acre for housing, retail, or industry.
- You have limited income this year. If you spend $17 on a music CD, that's $17 fewer you have to spend on other products or to save.

People produce goods (music CDs, houses, and parks) and services (the advice of physicians and lawyers) by using one or more of the following five **factors of production**, also called *production inputs* or simply *resources*:

factors of production
The resources used to produce goods and services; also known as *production inputs* or *resources*.

- **Natural resources** are provided by nature. Some examples are fertile land, mineral deposits, oil and gas deposits, and water. Some economists refer to all types of natural resources as *land*.
- **Labor** is the physical and mental effort people use to produce goods and services.
- **Physical capital** is the stock of equipment, machines, structures, and infrastructure that is used to produce goods and services. Some examples are forklifts, machine tools, computers, factories, airports, roads, and fiber-optic cables.
- **Human capital** is the knowledge and skills acquired by a worker through education and experience. Every job requires some human capital: To be a surgeon, you must learn anatomy and acquire surgical skills. To be an accountant, you must learn the rules of accounting and acquire computer skills. To be a musician, you must learn to play an instrument.
- **Entrepreneurship** is the effort used to coordinate the factors of production—natural resources, labor, physical capital, and human capital—to produce and sell products. An entrepreneur comes up with an idea for a product, decides how to produce it, and raises the funds to bring it to the market. Some examples of entrepreneurs are Bill Gates of Microsoft, Steve Jobs of Apple Computer, Howard Schultz of Starbucks, and Ray Kroc of McDonald's.

natural resources
Resources provided by nature and used to produce goods and services.

labor
Human effort, including both physical and mental effort, used to produce goods and services.

physical capital
The stock of equipment, machines, structures, and infrastructure that is used to produce goods and services.

human capital
The knowledge and skills acquired by a worker through education and experience and used to produce goods and services.

entrepreneurship
The effort used to coordinate the factors of production—natural resources, labor, physical capital, and human capital—to produce and sell products.

Given our limited resources, we make our choices in a variety of ways. Sometimes we make our decisions as individuals, and other times we participate in collective decision making, allowing the government and other organizations to choose for us. Many of our choices happen within *markets*, institutions or arrangements that enable us to buy and sell things. For example, most of us participate in the labor market, exchanging our time for money, and we all participate in consumer markets, exchanging money for food and clothing. But we make other choices outside markets—from our personal decisions about everyday life to our political choices about matters that concern society as a whole. What unites all these decisions is the notion of scarcity: We can't have it all; there are trade-offs.

Economists are always reminding us that there is scarcity—there are trade-offs in everything we do. Suppose that in a conversation with your economics instructor you share your enthusiasm about an upcoming launch of the space shuttle. The economist may tell you that the resources used for the shuttle could have been used instead for an unmanned mission to Mars.

By introducing the notion of scarcity into your conversation, your instructor is simply reminding you that there are trade-offs, that one thing (a Mars mission) is sacrificed for another (a shuttle mission). Talking about alternatives is the first step in a process that can help us make better choices about how to use our resources. For example, we could compare the scientific benefits of a shuttle mission to the benefits of a Mars mission and choose the mission with the greater benefit.

Positive versus Normative Analysis

Economics doesn't tell us what to choose—shuttle mission or Mars mission—but simply helps us to understand the trade-offs. President Harry S. Truman once remarked,

> All my economists say, "On the one hand, . . .; On the other hand, . . .?." Give me a one-handed economist!

An economist might say, "On the one hand, we could use a shuttle mission to do more experiments in the gravity-free environment of Earth's orbit; on the other hand, we could use a Mars mission to explore the possibility of life on other planets." In using both hands, the economist is not being evasive, but simply doing economics, discussing the alternative uses of our resources. The ultimate decision about how to use our resources—shuttle mission or Mars exploration—is the responsibility of citizens or their elected officials.

positive analysis
Answers the question "What *is*?" or "What *will be*?"

normative analysis
Answers the question "What *ought to be*?"

Most modern economics is based on **positive analysis**, which predicts the consequences of alternative actions by answering the question "What *is*?" or "What *will be*?" A second type of economic reasoning is normative in nature. **Normative analysis** answers the question "What *ought to be*?"

In Table 1.1, we compare positive questions to normative questions. Normative questions lie at the heart of policy debates. Economists contribute to policy debates by conducting positive analyses of the consequences of alternative actions. For example, an economist could predict the effects of an increase in the minimum wage on the number of people employed nationwide, the income of families with minimum-wage workers, and consumer prices. Armed with the conclusions of the economist's positive analysis, citizens and policymakers could then make a normative decision about whether to increase the minimum wage. Similarly, an economist could study the projects that could be funded with $1 billion in foreign aid, predicting the effects of each project on the income per person in an African country. Armed with this positive analysis, policymakers could then decide which projects to support.

TABLE 1.1 Comparing Positive and Normative Questions

Positive Questions	Normative Questions
• If the government increases the minimum wage, how many workers will lose their jobs?	• Should the government increase the minimum wage?
• If two office-supply firms merge, will the price of office supplies increase?	• Should the government block the merger of two office-supply firms?
• How does a college education affect a person's productivity and earnings?	• Should the government subsidize a college education?
• How do consumers respond to a cut in income taxes?	• Should the government cut taxes to stimulate the economy?
• If a nation restricts shoe imports, who benefits and who bears the cost?	• Should the government restrict imports?

Economists don't always reach the same conclusions in their positive analyses. The disagreements often concern the magnitude of a particular effect. For example, most economists agree that an increase in the minimum wage will cause unemployment, but disagree about how many people would lose their jobs. Similarly, economists agree that spending money to improve the education system in Africa will increase productivity and income, but disagree about the size of the increase in income. MyLab Economics Concept Check

The Three Key Economic Questions: What, How, and Who?

We make economic decisions at every level in society. Individuals decide what products to buy, what occupations to pursue, and how much money to save. Firms decide what goods and services to produce and how to produce them. Governments decide what projects and programs to complete and how to pay for them. The choices of individuals, firms, and governments answer three questions:

1. *What products do we produce?* Trade-offs exist: If a hospital uses its resources to perform more heart transplants, it has fewer resources to care for premature infants.
2. *How do we produce the products?* Alternative means of production are available: Power companies can produce electricity with coal, natural gas, or wind power. Professors can teach in large lecture halls or small classrooms.
3. *Who consumes the products?* We must decide how to distribute the products of society. If some people earn more money than others, should they consume more goods? How much money should the government take from the rich and give to the poor?

As we'll see later in the book, most of these decisions are made in markets, where prices play a key role in determining what products we produce, how we produce them, and who gets the products. In Chapter 3, we examine the role of markets in modern economies and the role of government in market-based economies. MyLab Economics Concept Check

Economic Models

Economists use *economic models* to explore the choices people make and the consequences of those choices. An economic model is a simplified representation of an economic environment, with all but the essential features of the environment eliminated. An **economic model** is an abstraction from reality that enables us to focus our attention on what really matters. As we'll see throughout the book, most economic models use graphs to represent the economic environment.

economic model
A simplified representation of an economic environment, often employing a graph.

To see the rationale for economic modeling, consider an architectural model. An architect builds a scale model of a new building and uses the model to show how the building will fit on a plot of land and blend with nearby buildings. The model shows the exterior features of the building, but not the interior features. We can ignore the interior features because they are unimportant for the task at hand—seeing how the building will fit into the local environment.

Economists build models to explore decision making by individuals, firms, and other organizations. For example, we can use a model of a profit-maximizing firm to predict how a firm will respond to increased competition. If a new car stereo store opens up in your town, will the old firms be passive and simply accept smaller market shares, or will they aggressively cut their prices to try to drive the new rival out of business? The model of the firm includes the monetary benefits and costs of doing business, and assumes that firms want to make as much money as possible. Although there may be other motives in the business world—to have fun or to help the world—the economic model ignores these other motives. The model focuses our attention on the profit motive and how it affects a firm's response to increased competition. MyLab Economics Concept Check

MyLab Economics Study Plan

Learning Objective 1.2

Discuss the insights from economics for a real-world problem such as congestion.

Economic Analysis and Modern Problems

Economic analysis provides important insights into real-world problems. To explain how we can use economic analysis in problem solving, we provide three examples. You'll see these examples again in more detail later in the book.

Economic View of Traffic Congestion

Consider first the problem of traffic congestion. According to the Texas Transportation Institute, the typical U.S. commuter wastes about 47 hours per year because of traffic congestion.[1] In some cities, the time wasted is much greater: 93 hours in Los Angeles, 72 hours in San Francisco, and 63 hours in Houston. In addition to time lost, we also waste 2.3 billion gallons of gasoline and diesel fuel each year.

To an economist, the diagnosis of the congestion problem is straightforward. When you drive onto a busy highway during rush hour, your car takes up space and decreases the distance between the vehicles on the highway. A driver's normal reaction to a shorter distance between moving cars is to slow down. So when you enter the highway, you force other commuters to slow down and thus spend more time on the highway. If each of your 900 fellow commuters spends just two extra seconds on the highway, you will increase the total travel time by 30 minutes. In deciding whether to use the highway, you will presumably ignore these costs you impose on others. Similarly, your fellow commuters ignore the cost they impose on you and others when they enter the highway. Because no single commuter pays the full cost (30 minutes), too many people use the highway, and everyone wastes time.

One possible solution to the congestion problem is to force people to pay for using the road, just as they pay for gasoline and tires. The government could impose a congestion tax of $8 per trip on rush-hour commuters and use a debit card system to collect the tax: Every time a car passes a checkpoint, a transponder would charge the commuter's card. Traffic volume during rush hours would then decrease as travelers (a) shift their travel to off-peak times, (b) switch to ride-sharing and mass transit, and (c) shift their travel to less congested routes. The job for the economist is to compute the appropriate congestion tax and predict the consequences of imposing it. MyLab Economics Concept Check

Economic View of Poverty in Africa

Consider next the issue of poverty in Africa. In the final two decades of the twentieth century, the world economy grew rapidly, and the average per capita income (income per person) increased by about 35 percent. In contrast, the economies of poverty-stricken sub-Saharan Africa shrank, and per capita income *decreased* by about 6 percent. Africa is the world's second-largest continent in both area and population and accounts for more than 12 percent of the world's human population. Figure 1.1 shows a map of Africa. The countries of sub-Saharan Africa are highlighted in orange.

Economists have found that as a nation's economy grows, its poorest households share in the general prosperity.[2] Therefore, one way to reduce poverty in sub-Saharan Africa is to increase economic growth. Economic growth occurs when a country expands its production facilities (machinery and factories), improves its public infrastructure (highways and water systems), widens educational opportunities, and adopts new technology.

The recent experience of sub-Saharan Africa is somewhat puzzling because in the last few decades the region has expanded educational opportunities and received large amounts of foreign aid. Some recent work by economists on the sources of growth suggests that institutions such as the legal system and the regulatory environment also play key roles in economic growth.[3] In sub-Saharan Africa, a simple legal dispute about a small debt takes about 30 months to resolve, compared to 5 months in the United States. In Mozambique, it takes 174 days to complete the procedures required to set up a business, compared to just 2 days in Canada. In many cases, institutions impede

▲ **FIGURE 1.1** Map of Africa

Africa is the world's second-largest continent in both area and population, and accounts for more than 12 percent of the world's human population. The countries of sub-Saharan Africa are highlighted in green.

rather than encourage the sort of investment and risk-taking—called *entrepreneurship*—that causes economic growth and reduces poverty. As a consequence, economists and policymakers are exploring ways to reform the region's institutions. They are also challenged with choosing among development projects that will generate the biggest economic boost per dollar spent—the biggest bang per buck. MyLab Economics Concept Check

Economic View of the Current World Recession

Over the last several decades, the U.S. economy has performed well and has raised our standard of living. The general consensus was that our policymakers had learned to manage the economy effectively. Although the economy faltered at times, policymakers seemed to know how to restore growth and prosperity.

That is why the financial crisis and the recession that began in late 2007 has so shaken the confidence of people in the United States and around the world. The problems started innocently enough, with a booming market for homes that was fueled by easy credit from financial institutions. But we later discovered that many purchasers of homes and properties could not really afford them, and when many homeowners had trouble making their mortgage payments, the trouble spread to banks and other financial institutions. As a result, businesses found it increasingly difficult to borrow money for everyday use and investment, and economic activity around the world began to contract.

The major countries of the world have implemented aggressive policies to try to halt this downturn. Policymakers want to avoid the catastrophes that hit the global economy in the 1930s. Fortunately, they can draw on many years of experience in economic policy to guide the economy during this difficult time. MyLab Economics Concept Check

MyLab Economics Study Plan

The Economic Way of Thinking

Learning Objective 1.3

List the four elements of the economic way of thinking.

How do economists think about problems and decision making? The economic way of thinking is best summarized by British economist John Maynard Keynes (1883–1946):[4]

> The theory of economics does not furnish a body of settled conclusions immediately applicable to policy. It is a method rather than a doctrine, an apparatus of the mind, a technique of thinking which helps its possessor draw correct conclusions.

Let's look at the four elements of the economic way of thinking.

Use Assumptions to Simplify

Economists use assumptions to make things simpler and focus attention on what really matters. If you use a road map to plan a car trip from Seattle to San Francisco, you make two unrealistic assumptions to simplify your planning:

- The earth is flat: The flat road map doesn't show the curvature of the earth.
- The roads are flat: The standard road map doesn't show hills and valleys.

Instead of a map, you could use a globe that shows all the topographical features between Seattle and San Francisco, but you don't need those details to plan your trip. A map, with its unrealistic assumptions, will suffice because the curvature of the earth and the topography of the highways are irrelevant to your trip. Although your analysis is based on two unrealistic assumptions, that does not mean your analysis is invalid. Similarly, if economic analysis is based on unrealistic assumptions, that doesn't mean the analysis is faulty.

What if you decide to travel by bike instead of by automobile? Now the assumption of flat roads really matters, unless of course you are eager to pedal up and down mountains. If you use a standard map, and thus assume there are no mountains between the two cities, you may inadvertently pick a mountainous route instead of a flat one. In this case, the simplifying assumption makes a difference. The lesson is that we must think carefully about whether a simplifying assumption is truly harmless. MyLab Economics Concept Check

Isolate Variables—*Ceteris Paribus*

variable
A measure of something that can take on different values.

Economic analysis often involves variables and how they affect one another. A **variable** is a measure of something that can take on different values, for example, your grade point average. Economists are interested in exploring relationships between two variables—like the relationship between the price of apples and the quantity of apples consumers purchase. Of course, the quantity of apples purchased depends on many other variables, including the consumer's income. To explore the relationship between the quantity and price of apples, we must assume that the consumer's income—and anything else that influences apple purchases—doesn't change during the time period we're considering.

Alfred Marshall (1842–1924) was a British economist who refined the economic model of supply and demand and provided a label for this process.[5] He picked one variable that affected apple purchases (price) and threw the other variable (income) into what he called the "pound" (in Marshall's time, the "pound" was an enclosure for holding stray cattle; nowadays, a pound is for stray dogs). That variable waited in the pound while Marshall examined the influence of the first variable. Marshall labeled the pound ***ceteris paribus***, the Latin expression meaning that other variables are held fixed:

ceteris paribus
The Latin expression meaning that other variables are held fixed.

> . . . the existence of other tendencies is not denied, but their disturbing effect is neglected for a time. The more the issue is narrowed, the more exactly can it be handled.

This text contains many statements about the relationship between two variables. For example, the quantity of computers produced by Dell depends on the price of computers, the wage of computer workers, and the cost of microchips. When we say, "An increase in the price of computers increases the quantity of computers produced," we are assuming that the other two variables—the wage and the cost of microchips—do not change. That is, we apply the *ceteris paribus* assumption. MyLab Economics Concept Check

Think at the Margin

marginal change
A small, one-unit change in value.

Economists often consider how a small change in one variable affects another variable and what impact that has on people's decision making. In other words, if circumstances change only slightly, how will people respond? A small, one-unit change in value is called a **marginal change**. The key feature of marginal change is that the first variable changes by only one unit. For example, you might ask, "If I study just one more hour, by how much will my exam score increase?" Economists call this process "thinking at the margin." Thinking at the margin is like thinking on the edge. You will encounter marginal thinking throughout this text. Here are some other marginal questions:

- If I keep my barber shop open one more hour, by how much will my revenue increase?
- If I stay in school and earn another degree, by how much will my lifetime earnings increase?
- If a car dealer hires one more sales associate, how many more cars will the dealer sell?

As we'll see in the next chapter, economists use the answer to a marginal question as a first step in deciding whether to do more or less of something, for example, whether to keep your barber shop open one more hour. MyLab Economics Concept Check

Rational People Respond to Incentives

A key assumption of most economic analysis is that people act rationally, meaning they act in their own self-interest. Scottish philosopher Adam Smith (1723–1790), who is also considered the founder of economics, wrote that he discovered within humankind[6]

> a desire of bettering our condition, a desire which, though generally calm and dispassionate, comes with us from the womb, and never leaves us until we go to the grave.

Smith didn't say people are motivated exclusively by self-interest, but rather that self-interest is more powerful than kindness or altruism. In this text, we will assume that people act in their own self-interest. Rational people respond to incentives. When the payoff, or benefit, from doing something changes, people change their behavior to get the benefit. MyLab Economics Concept Check

Application 1

INCENTIVES TO BUY HYBRID VEHICLES

APPLYING THE CONCEPTS #1: How do people respond to incentives?

Consider the incentives to buy a hybrid vehicle, which is more fuel efficient but more expensive than a gas-powered vehicle. Between 2000 and 2007, the number of hybrid vehicles increased from fewer than 10,000 vehicles to more than 340,000 vehicles. Over this period, the price of gasoline increased significantly, and the higher price of gasoline was responsible for roughly one-third of the hybrid vehicles purchased in 2007. An additional factor in hybrid purchases was a federal subsidy of up to $3,400 per hybrid vehicle. The subsidy was responsible for roughly one-fifth of the hybrid vehicles purchased in 2007. The increase in the number of hybrid vehicles decreased the emission of the greenhouse gas carbon dioxide (CO_2).

How efficient is the hybrid subsidy in reducing CO_2? On average, the cost of abating one ton of CO_2 through the hybrid subsidy is $177. There are less costly ways to reduce CO_2 emissions, including building insulation, energy-efficient lighting, reforestation, and switching to electric power systems that use fuels that generate less CO_2. For example, a switch from coal to natural gas in power plants reduces CO_2 emissions at less than one-third the cost associated with the hybrid subsidy. **Related to Exercise 3.4.**

SOURCES: (1) Arie Beresteanu and Shanjun Li, "Gasoline Prices, Government Support, and the Demand for Hybrid Vehicles in the United States," *International Economic Review* 52 (2011), pp. 161–182. (2) Jackie Calmes, "President Pushes to Add More Credits for Hybrids," *New York Times*, March 8, 2012.

Example: London Addresses Its Congestion Problem

To illustrate the economic way of thinking, let's consider again how an economist would approach the problem of traffic congestion. Recall that each driver on the highway slows down other drivers but ignores these time costs when deciding whether to use the highway. If the government imposes a congestion tax to reduce traffic during rush hour, the economist is faced with a question: How high should the tax be?

To determine the appropriate congestion tax, an economist would assume that people respond to incentives and use the three other elements of the economic way of thinking:

- **Use assumptions to simplify.** To simplify the problem, we would assume that every car has the same effect on the travel time of other cars. Of course, this is unrealistic because people drive cars of different sizes in different ways. But the alternative—looking at the effects of each car on travel speeds—would needlessly complicate the analysis.
- **Isolate variables—use *ceteris paribus*.** To focus attention on the effects of a congestion tax on the number of cars using the highway, we would make the *ceteris paribus* assumption that everything else that affects travel behavior—the price of gasoline, bus fares, and consumer income—remains fixed.
- **Think at the margin.** To think at the margin, we would estimate the effects of adding one more car to the highway. Now consider the marginal question: If we add one more car to the highway, by how much does the total travel time for commuters increase? Once we answer this question, we can determine the cost imposed by the marginal driver. If the marginal driver forces each of the 900 commuters to spend two extra seconds on the highway, total travel time increases by 30 minutes. If the value of time is, say, \$16 per hour, the appropriate congestion tax would be \$8 (equal to \$16 × 1/2 hour).

Application 2

HOUSING PRICES IN CUBA

APPLYING THE CONCEPTS #2: What is the role of prices in allocating resources?

As an illustration of the role of prices in an economy, consider the experience of Cuba without housing prices. In 1960, the government confiscated most housing and outlawed its sale and rental. Tenants who paid monthly rent to the government for 20 years became homeowners of sorts—they officially owned the dwelling, but could not sell it or rent it to others. The impossibility of resale and rental meant that homeowners had less incentive to repair and maintain their property. As a result, a large fraction of the housing stock is in mediocre or poor condition (30 percent in 2013). In addition, few new dwellings were built in the last 50 years, leading to a large housing deficit (roughly 1 million dwellings in 2013).

Housing reforms in 2011 restored prices to the Cuban housing market. The reforms authorize the sale and purchase of homes for Cuban citizens at prices chosen by sellers and buyers. Economists expect these reforms to provide homeowners with greater incentives to repair and maintain their dwellings, and to increase housing construction. **Related to Exercise 3.5**.

SOURCES: (1) Carmelo Mesa-Lago, *Institutional Changes of Cuba's Economic Social Reforms: State and Market Roles, Progress, Hurdles, Comparisons, Monitoring and Effects* (Washington, DC: Brookings, 2014). (2) Damien Cave, "Cuba to Allow Buying and Selling of Property, With Few Restrictions," *New York Times*, November 3, 2011.

If the idea of charging people for using roads seems odd, consider the city of London, which for decades had experienced the worst congestion in Europe. In February 2003, the city imposed an $8 tax per day to drive in the city between 7:00 A.M. and 6:30 P.M. The tax reduced traffic volume and cut travel times for cars and buses in half. Because the tax reduced the waste and inefficiency of congestion, the city's economy thrived. Given the success of London's ongoing congestion tax, other cities, including Toronto, Singapore, and San Diego, have implemented congestion pricing. MyLab Economics Concept Check

MyLab Economics Study Plan

Preview of Coming Attractions: Macroeconomics

Learning Objective 1.4

List three ways to use macroeconomics.

The field of economics is divided into two categories: macroeconomics and microeconomics. **Macroeconomics** is the study of the nation's economy as a whole; it focuses on the issues of inflation (a general rise in prices), unemployment, and economic growth. These issues are regularly discussed on Web sites, in newspapers, and on television. Macroeconomics explains why economies grow and change and why economic growth is sometimes interrupted. Let's look at three ways we can use macroeconomics.

macroeconomics

The study of the nation's economy as a whole; focuses on the issues of inflation, unemployment, and economic growth.

Using Macroeconomics to Understand Why Economies Grow

As we discussed earlier in the chapter, the world economy has been growing in recent decades, with per capita income increasing by about 1.5 percent per year. Increases in income translate into a higher standard of living for consumers—better cars, houses, and clothing and more options for food, entertainment, and travel. People in a growing economy can consume more of all goods and services because the economy has more of the resources needed to produce these products. Macroeconomics explains why resources increase over time and the consequences for our standard of living. Let's look at a practical question about economic growth.

Why do some countries grow much faster than others? Between 1960 and 2001, the economic growth rate was 2.2 percent per year in the United States, compared to 2.3 percent in Mexico and 2.7 percent in France. But in some countries, the economy actually shrunk, and per capita income dropped. Among the countries with declining income were Sierra Leone and Haiti. In the fastest-growing countries, citizens save a large fraction of the money they earn. Firms can then borrow the funds saved to purchase machinery and equipment that make their workers more productive. The fastest-growing countries also have well-educated workforces, allowing firms to quickly adopt new technologies that increase worker productivity. MyLab Economics Concept Check

Using Macroeconomics to Understand Economic Fluctuations

All economies, including those that experience a general trend of rising per capita income, are subject to economic fluctuations, including periods when the economy temporarily shrinks. During an economic downturn, some of the economy's resources—natural resources, labor, physical capital, human capital, and entrepreneurship—are idle. Some workers are unemployed, and some factories and stores are closed. By contrast, sometimes the economy grows too rapidly, causing prices to rise. Macroeconomics helps us understand why these fluctuations occur—why the economy sometimes cools and sometimes overheats—and what the government can do to moderate the fluctuations. Let's look at a practical question about economic fluctuations.

Should Congress and the president do something to reduce the unemployment rate? For example, should the government cut taxes to free up income to spend on consumer goods, thus encouraging firms to hire more workers to produce more output? If unemployment is very high, the government may want to reduce it. However, it is important not to reduce the unemployment rate too much because, as we'll see later in the book, a low unemployment rate will cause inflation. MyLab Economics Concept Check

Using Macroeconomics to Make Informed Business Decisions

A third reason for studying macroeconomics is to make informed business decisions. As we'll see later in the book, the government uses various policies to influence interest rates (the price of borrowing money) and the inflation rate. A manager who intends to borrow money for a new factory or store could use knowledge of macroeconomics to predict the effects of current public policies on interest rates and then decide whether to borrow the money now or later. Similarly, a manager must keep an eye on the inflation rate to help decide how much to charge for the firm's products and how much to pay workers. A manager who studies macroeconomics will be better equipped to understand the complexities of interest rates and inflation and how they affect the firm. MyLab Economics Concept Check

MyLab Economics Study Plan

Learning Objective 1.5

List three ways to use microeconomics.

microeconomics
The study of the choices made by households, firms, and government and how these choices affect the markets for goods and services.

Preview of Coming Attractions: Microeconomics

Microeconomics is the study of the choices made by households (an individual or a group of people living together), firms, and government and how these choices affect the markets for goods and services. Let's look at three ways we can use microeconomic analysis.

Using Microeconomics to Understand Markets and Predict Changes

One reason for studying microeconomics is to better understand how markets work and to predict how various events affect the prices and quantities of products in markets. In this text, we answer many practical questions about markets and how they operate. Let's look at a practical question that can be answered with some simple economic analysis.

How would a tax on beer affect the number of highway deaths among young adults? Research has shown that the number of highway fatalities among young adults is roughly proportional to the total amount of beer consumed by that group. A tax on beer would make the product more expensive, and young adults, like other beer drinkers, would therefore consume less of it. Consequently, a tax that decreases beer consumption by 10 percent will decrease highway deaths among young adults by about 10 percent, too. MyLab Economics Concept Check

Using Microeconomics to Make Personal and Managerial Decisions

On the personal level, we use economic analysis to decide how to spend our time, what career to pursue, and how to spend and save the money we earn. Managers use economic analysis to decide how to produce goods and services, how much to produce, and how much to charge for them. Let's use some economic analysis to look at a practical question confronting someone considering starting a business.

If the existing coffee shops in your city are profitable and you have enough money to start your own shop, should you do it? If you enter this market, the competition among the shops for consumers will heat up, causing some coffee shops to drop their prices. In addition, your costs may be higher than the costs of the stores that are already established. It would be sensible to enter the market only if you expect a small drop in price and a small difference in costs. Indeed, entering what appears to be a lucrative market may turn out to be a financial disaster. MyLab Economics Concept Check

Using Microeconomics to Evaluate Public Policies

Although modern societies use markets to make most of the decisions about production and consumption, the government does fulfill several important roles. We can use economic analysis to determine how well the government performs its roles in the market economy. We can also explore the trade-offs associated with various public policies. Let's look at a practical question about public policy.

Like other innovations, prescription drugs are protected by government patents, giving the developer the exclusive right to sell a new drug for a fixed period of time. Once the patent expires, other pharmaceutical companies can legally produce and sell generic versions of the drug, which causes its price to drop. Should drug patents be shorter? Shortening the patent has trade-offs. The good news is that generic versions of the drug will be available sooner, so prices will drop sooner and more people will use the drug to improve their health. The bad news is that the financial payoff from developing new drugs will be smaller, so pharmaceutical companies won't develop as many new drugs. The question is whether the benefit of shorter patents (lower prices) exceeds the cost (fewer drugs developed). MyLab Economics Concept Check

MyLab Economics Study Plan

Chapter Summary and Problems

SUMMARY

Economics is about making choices when options are limited. Options in an economy are limited because the factors of production are limited. We can use economic analysis to understand the consequences of our choices as individuals, organizations, and society as a whole. Here are the main points of the chapter:

1. Most of modern economics is based on *positive analysis*, which answers the question "What *is*?" or "What *will be*?" Economists contribute to policy debates by conducting positive analyses about the consequences of alternative actions.
2. *Normative analysis* answers the question "What *ought to be*?"
3. The choices made by individuals, firms, and governments answer three questions: What products do we produce? How do we produce the products? Who consumes the products?
4. To think like an economist, we (a) use assumptions to simplify, (b) use the notion of *ceteris paribus* to focus on the relationship between two variables, (c) think in marginal terms, and (d) assume that rational people respond to incentives.
5. We use *macroeconomics* to understand why economies grow, to understand economic fluctuations, and to make informed business decisions.
6. We use *microeconomics* to understand how markets work, to make personal and managerial decisions, and to evaluate the merits of public policies.

KEY TERMS

ceteris paribus, p. 38
economic model, p. 35
economics, p. 33
entrepreneurship, p. 33
factors of production, p. 33
human capital, p. 33
labor, p. 33
macroeconomics, p. 41
marginal change, p. 38
microeconomics, p. 42
natural resources, p. 33
normative analysis, p. 34
physical capital, p. 33
positive analysis, p. 34
scarcity, p. 33
variable, p. 38

EXERCISES

All problems are assignable in MyLab Economics.

What Is Economics?

List the three key economic questions.

1.1 The three basic economic questions a society must answer are _________ products do we produce? _________ do we produce the products? _________ consumes the products?

1.2 List the five factors of production.

1.3 Which of the following statements is true?

a. Positive statements answer questions like "What will happen if . . .?" Normative economic statements answer questions like "What ought to happen to . . .?"

b. Normative statements answer questions like "What will happen if . . .?" Positive economic statements answer questions like "What ought to happen to . . .?"

c. Most modern economics is based on normative analysis.

1.4 Indicate whether the each of the following questions is normative or positive:

a. In the case of normal goods, will an increase in price lead to a decrease in quantity demanded?

b. Should the government provide homes for all citizens?

c. Is the rate of unemployment in Poland lower than the rate of unemployment in France?

d. How did the flood of 1997 affect housing prices in Wroclaw?

e. Should the rich be taxed at a far higher rate than the poor?

Economic Analysis and Modern Problems

Discuss the insights from economics for a real-world problem such as congestion.

2.1 What is the economist's solution to the problem of smog in cities?

a. Ban the use of cars in cities.

b. Require commuters to use bicycles at least twice a week.

c. Tax the owners of vehicles that produce significant amounts of air pollution.

d. No economist would suggest any of the above.

2.2 Some recent work by economists on the sources of growth suggests that institutions such as the _________ and the _________ play key roles in economic growth.

The Economic Way of Thinking

List the four elements of the economic way of thinking.

3.1 A road map incorporates two unrealistic assumptions: (1) _________ and (2) _________.

3.2 The four elements of the economic way of thinking are (1) use _________ to simplify the analysis, (2) explore the relationship between two variables by _________, (3) think at the _________, and (4) rational people respond to _________.

3.3 Which of the following is the Latin expression meaning *other things being held fixed*?

a. *setiferous proboscis*

b. *ceteris paribus*

c. *e pluribus unum*

d. *tres grand fromage*

3.4 Between 2000 and 2007, the number of hybrid vehicles increased from fewer than 10,000 to more than _________. One of the reasons of the increased hybrid purchases was a federal subsidy of up to $3,400 per hybrid vehicle. (Related to Application 1 on page 39.)

3.5 The inability of Cuban homeowners to sell their dwellings led to less investment in ___________. (Related to Application 2 on page 40.)

3.6 True or False: Adam Smith suggested that people are motivated solely by self-interest.

APPENDIX

USING GRAPHS AND PERCENTAGES

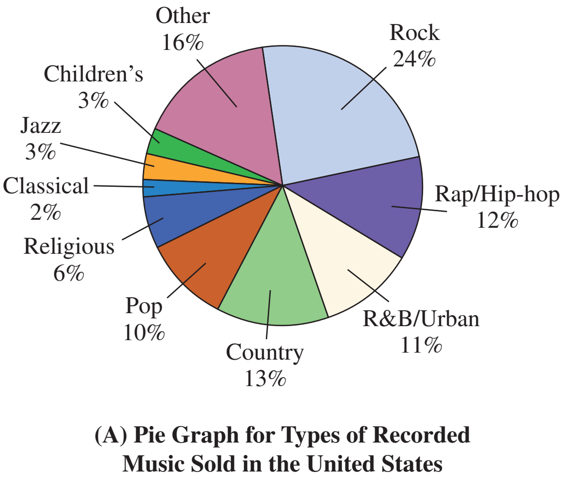
Economists use several types of graphs to present data, represent relationships between variables, and explain concepts. In this appendix, we review the mechanics of graphing variables. We also review the basics of computing percentage changes and using percentages to compute changes in variables.

1A.1 Using Graphs

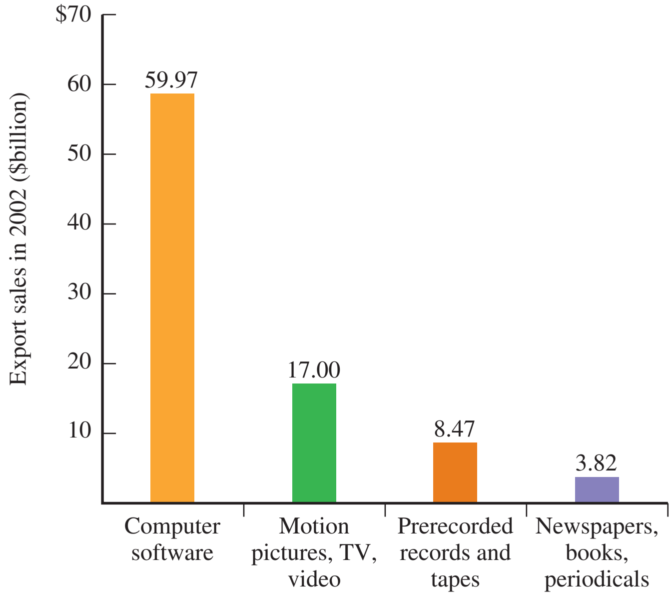
A quick flip through the book will reveal the importance of graphs in economics. Every chapter has at least several graphs, and many chapters have more. Although it is possible to do economics without graphs, it's a lot easier with them in your toolbox.

Graphing Single Variables

As we saw earlier in this chapter, a *variable* is a measure of something that can take on different values. Figure 1A.1 shows two types of graphs, each presenting data on a single variable. Panel A uses a pie graph to show the breakdown of U.S. music sales by type of music. The greater the sales of a type of music, the larger the pie slice. For example, the most popular type is rock music, comprising 24 percent of the market. The next largest type is country, followed by rap/hip-hop, R&B/urban, and so on. Panel B of Figure 1A.1 uses a bar graph to show the revenue from foreign sales (exports) of selected U.S. industries. The larger the revenue, the taller the bar. For example, the bar for computer software, with export sales of about $60 billion, is over three times taller than the bar for motion pictures, TV, and video, with export sales of $17 billion.



(A) Pie Graph for Types of Recorded Music Sold in the United States



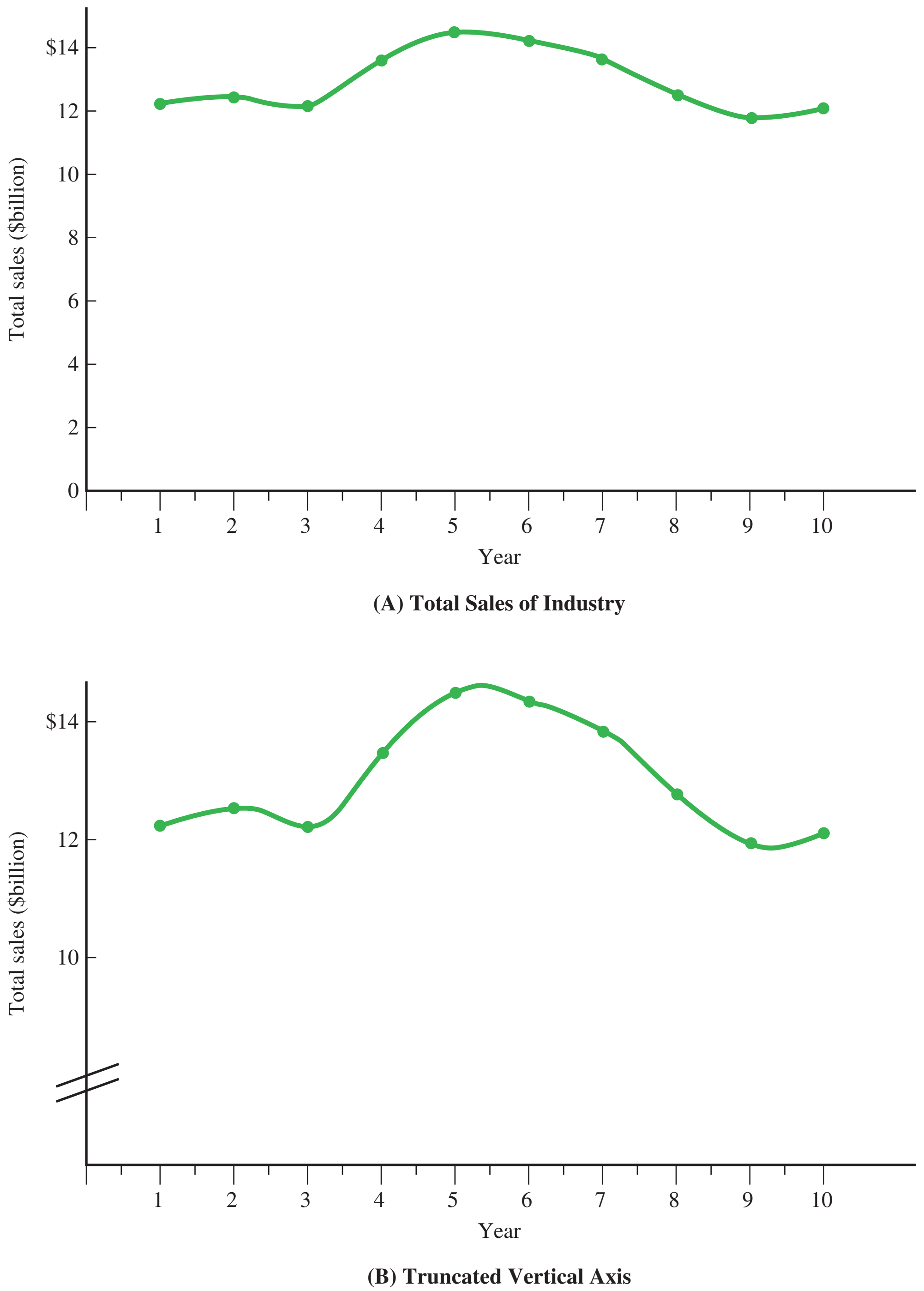
(B) Bar Graph for U.S. Export Sales of Copyrighted Products

▲ **FIGURE 1A.1** Graphs of Single Variables

SOURCE: Author's calculations based on Recording Industry Association of America, "2004 Consumer Profile."

SOURCE: Author's calculations based on International Intellectual Property Alliance, "Copyright Industries in the U.S. Economy, 2004 Report."

A third type of single-variable graph shows how the value of a variable changes over time. Panel A of Figure 1A.2 shows a time-series graph, with the total dollar value of a hypothetical industry for years 1 through 10. Time is measured on the horizontal axis, and sales are measured on the vertical axis. The height of the line in a particular year shows the value in that year. For example, in Year 1 the value was $12.32 billion. After reaching a peak of $14.59 billion in Year 5, the value dropped over the next several years.

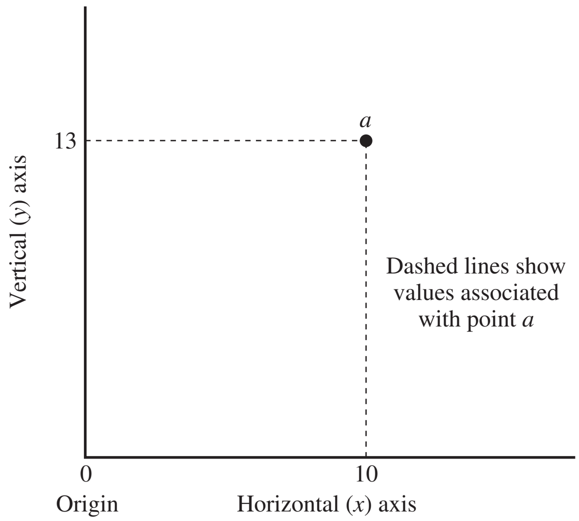


▲ **FIGURE 1A.2** Time-Series Graph

Panel B of Figure 1A.2 shows a truncated version of the graph in Panel A. The double hash marks in the lower part of the vertical axis indicate that the axis doesn't start from zero. This truncation of the vertical axis exaggerates the fluctuations in total sales. MyLab Economics Concept Check

Graphing Two Variables

We can also use a graph to show the relationship between two variables. Figure 1A.3 shows the basic elements of a two-variable graph. One variable is measured along the horizontal, or *x*, axis, while the other variable is measured along the vertical, or *y*, axis. The *origin* is the intersection of the two axes, where the values of both variables are zero. Dashed lines show the values of the two variables at a particular point. For example, for point *a*, the value of the horizontal, or *x*, variable is 10, and the value of the vertical, or *y*, variable is 13.

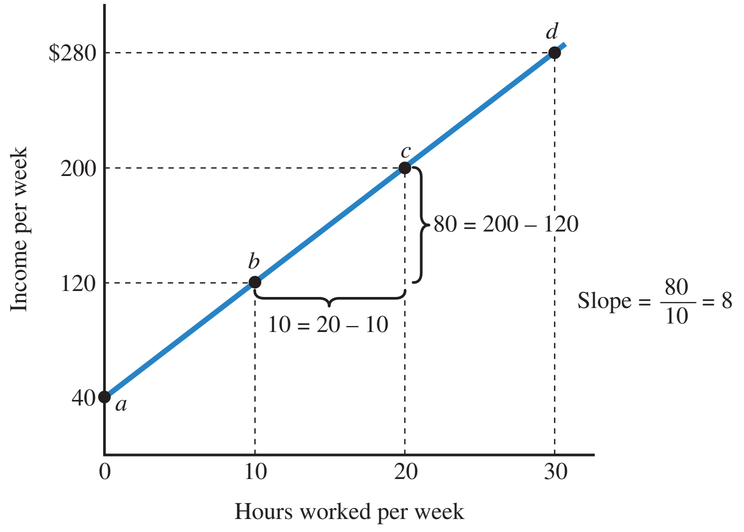


▲ **FIGURE 1A.3** Basic Elements of a Two-Variable Graph

One variable is measured along the horizontal, or *x*, axis, while the other variable is measured along the vertical, or *y*, axis. The origin is defined as the intersection of the two axes, where the values of both variables are zero. The dashed lines show the values of the two variables at a particular point.

To see how to draw a two-variable graph, suppose that you have a part-time job and you are interested in the relationship between the number of hours you work and your weekly income. The relevant variables are the hours of work per week and your weekly income. In Figure 1A.4, the table shows the relationship between the hours worked and income. Let's assume that your weekly allowance from your parents is $40 and your part-time job pays $8 per hour. If you work 10 hours per week, for example, your weekly income is $120 ($40 from your parents and $80 from your job). The more you work, the higher your weekly income: If you work 20 hours, your weekly income is $200; if you work 30 hours, it is $280.

Hours Worked per Week	Income per Week	Point on the Graph
0	$ 40	*a*
10	120	*b*
20	200	*c*
30	280	*d*



▲ **FIGURE 1A.4** Relationship between Hours Worked and Income

There is a positive relationship between work hours and income, so the income curve is positively sloped. The slope of the curve is \$8: Each additional hour of work increases income by \$8.

Although a table with numbers is helpful in showing the relationship between work hours and income, a graph makes it easier to see the relationship. We can use data in a table to draw a graph. To do so, we perform five simple steps:

1. Draw a horizontal line to represent the first variable. In Figure 1A.4, we measure hours worked along the horizontal axis. As we move to the right along the horizontal axis, the number of hours worked increases, from 0 to 30 hours.
2. Draw a vertical line intersecting the first line to represent the second variable. In Figure 1A.4, we measure income along the vertical axis. As we move up along the vertical axis, income increases from \$0 to \$280.
3. Start with the first row of numbers in the table, which shows that with 0 hours worked, income is \$40. The value of the variable on the horizontal axis is 0, and the value of the variable on the vertical axis is \$40, so we plot point *a* on the graph. This is the *vertical intercept*—the point where the curve cuts or intersects the vertical axis.
4. Pick a combination with a positive number for hours worked. For example, in the second row of numbers, if you work 10 hours, your income is \$120.
   4.1 Find the point on the horizontal axis with that number of hours worked—10 hours—and draw a dashed line vertically straight up from that point.
   4.2 Find the point on the vertical axis with the income corresponding to those hours worked—\$120—and draw a dashed line horizontally straight to the right from that point.
   4.3 The intersection of the dashed lines shows the combination of hours worked and income. Point *b* shows the combination of 10 hours worked and \$120 in income.
5. Repeat step 4 for different combinations of work time and income shown in the table. Once we have a series of points on the graph (*a*, *b*, *c*, and *d*), we can connect them to draw a curve that shows the relationship between hours worked and income.

There is a **positive relationship** between two variables if they move in the same direction. As you increase your work time, your income increases, so there is a positive relationship between the two variables. In Figure 1A.4, as the number of hours worked increases, you move upward along the curve to higher income levels. Some people refer to a positive relationship as a *direct relationship*.

positive relationship
A relationship in which two variables move in the same direction.

There is a **negative relationship** between two variables if they move in opposite directions. For example, there is a negative relationship between the amount of time you work and the time you have available for other activities such as recreation, study, and sleep. Some people refer to a negative relationship as an *inverse relationship*.

negative relationship
A relationship in which two variables move in opposite directions.

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Computing the Slope

How sensitive is one variable to changes in the other variable? We can use the slope of the curve to measure this sensitivity. To compute the **slope of a curve**, we pick two points and divide the vertical difference between the two points (the *rise*) by the horizontal difference (the *run*):

slope of a curve
The vertical difference between two points (the *rise*) divided by the horizontal difference (the *run*).

$$\text{Slope} = \frac{\text{Vertical difference between two points}}{\text{Horizontal difference between two points}} = \frac{\text{rise}}{\text{run}}$$

To compute the slope of a curve, we take four steps:

1. Pick two points on the curve, for example, points *b* and *c* in Figure 1A.4.
2. Compute the vertical difference between the two points (the rise). For points *b* and *c*, the vertical difference between the points is \$80 (\$200 − \$120).
3. Compute the horizontal distance between the same two points (the run). For points *b* and *c*, the horizontal distance between the points is 10 hours (20 hours − 10 hours).
4. Divide the vertical distance by the horizontal distance to get the slope. The slope between points *b* and *c* is \$8 per hour:

$$\text{Slope} = \frac{\text{Vertical difference}}{\text{Horizontal difference}} = \frac{\$200 - 120}{20 - 10} = \frac{\$80}{10} = \$8$$

In this case, a 10-hour increase in time worked increases income by \$80, so the increase in income per hour of work is \$8, which makes sense because this is the hourly wage. Because the curve is a straight line, the slope is the same at all points along the curve. You can check this yourself by computing the slope between points *c* and *d*.

We can use some shorthand to refer to the slope of a curve. The mathematical symbol Δ (delta) represents the change in a variable. So we can write the slope of the curve in Figure 1A.4 as

$$\text{Slope} = \frac{\Delta\ \text{Income}}{\Delta\ \text{Work hours}}$$

In general, if the variable on the vertical axis is *y* and the variable on the horizontal axis is *x*, we can express the slope as

$$\text{Slope} = \frac{\Delta y}{\Delta x}$$

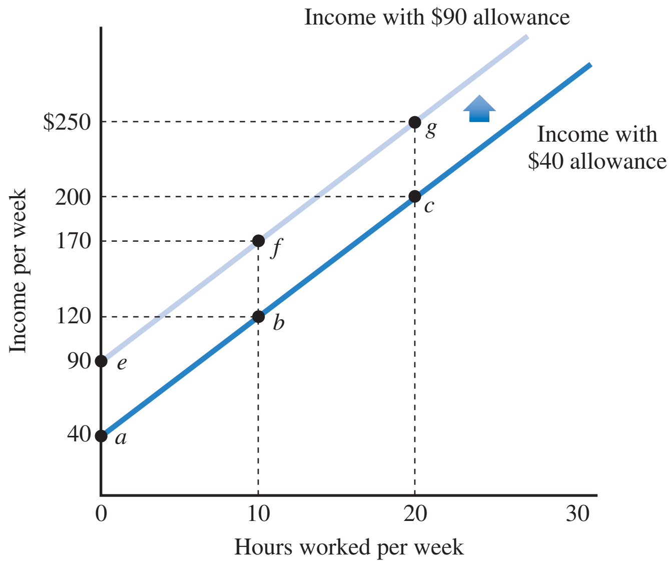
MyLab Economics Concept Check

Moving Along the Curve versus Shifting the Curve

Up to this point, we've explored the effect of changes in variables that cause movement along a given curve. In Figure 1A.4, we see the relationship between hours of work (on the horizontal axis) and income (on the vertical axis). Because the total income also depends on the allowance and the wage, we can make two observations about the curve in Figure 1A.4:

1. To draw this curve, we must specify the weekly allowance ($40) and the hourly wage ($8).
2. The curve shows that an increase in time worked increases the student's income, *ceteris paribus*. In this case, we are assuming that the allowance and the wage are both fixed.

A change in the weekly allowance will shift the curve showing the relationship between work time and income. In Figure 1A.5, when the allowance increases from $40 to $90, the curve shifts upward by $50. For a given number of work hours, income increases by $50. For example, the income associated with 10 hours of work is $170 (point *f*), compared to $120 with the original allowance (point *b*). The upward shift also means that to reach a given amount of income, fewer work hours are required. In other words, the curve shifts upward and to the left.



▲ **FIGURE 1A.5** Movement Along a Curve versus Shifting the Curve

To draw a curve showing the relationship between hours worked and income, we fix the weekly allowance ($40) and the wage ($8 per hour). A change in the hours worked causes movement along the curve, for example, from point *b* to point *c*. A change in any other variable shifts the entire curve. For example, a $50 increase in the allowance (to $90) shifts the entire curve upward by $50.

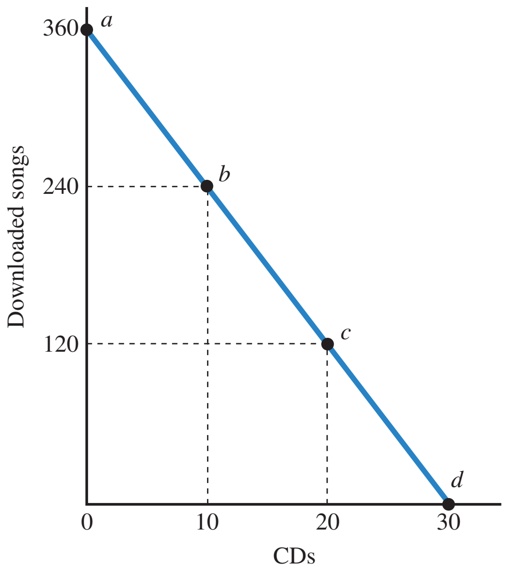
We can distinguish between movement along a curve and a shift of the entire curve. In Figure 1A.5, an increase in the hours worked causes movement along a single income curve. For example, if the allowance is $40, we are operating on the lower of the two curves, and if the hours worked increases from 10 to 20, we move from point *b* to point *c*. In contrast, if something other than the hours worked changes, we shift the entire curve, as we've seen with an increase in the allowance.

This book uses dozens of two-dimensional curves, each of which shows the relationship between *only two* variables. A common error is to forget that a single curve tells only part of the story. In Figure 1A.5, we needed two curves to explore the effects of changes in three variables. Here are some simple rules to keep in mind when you use two-dimensional graphs:

- A change in one of the variables shown on the graph causes movement along the curve. In Figure 1A.5, an increase in work time causes movement along the curve from point *a* to point *b*, to point *c*, and so on.
- A change in one of the variables that is not shown on the graph—one of the variables held fixed in drawing the curve—shifts the entire curve. In Figure 1A.5, an increase in the allowance shifts the entire curve upward. MyLab Economics Concept Check

Graphing Negative Relationships

We can use a graph to show a negative relationship between two variables. Consider a consumer who has an annual budget of $360 to spend on CDs at a price of $12 per CD and downloaded music at a price of $1 per song. The table in Figure 1A.6 shows the relationship between the number of CDs and downloaded songs. A consumer who doesn't buy any CDs has $360 to spend on downloaded songs and can get 360 of them at a price of $1 each. A consumer who buys 10 CDs at $12 each has $240 left to spend on downloaded songs (point *b*). Moving down through the table, as the number of CDs increases, the number of downloaded songs decreases.



▲ **FIGURE 1A.6** Negative Relationship between CD Purchases and Downloaded Songs

There is a negative relationship between the number of CDs and downloaded songs that a consumer can afford with a budget of $360. The slope of the curve is −$12: Each additional CD (at a price of $12 each) decreases the number of downloadable songs (at $1 each) by 12 songs.

Number of CDs Purchased	Number of Songs Downloaded	Point on the Graph
0	360	*a*
10	240	*b*
20	120	*c*
30	0	*d*

The graph in Figure 1A.6 shows the negative relationship between the number of CDs and the number of downloaded songs. The vertical intercept (point *a*) shows that a consumer who doesn't buy any CDs can afford 360 downloaded songs. There is a negative relationship between the number of CDs and downloaded songs, so the curve is negatively sloped. We can use points *b* and *c* to compute the slope of the curve:

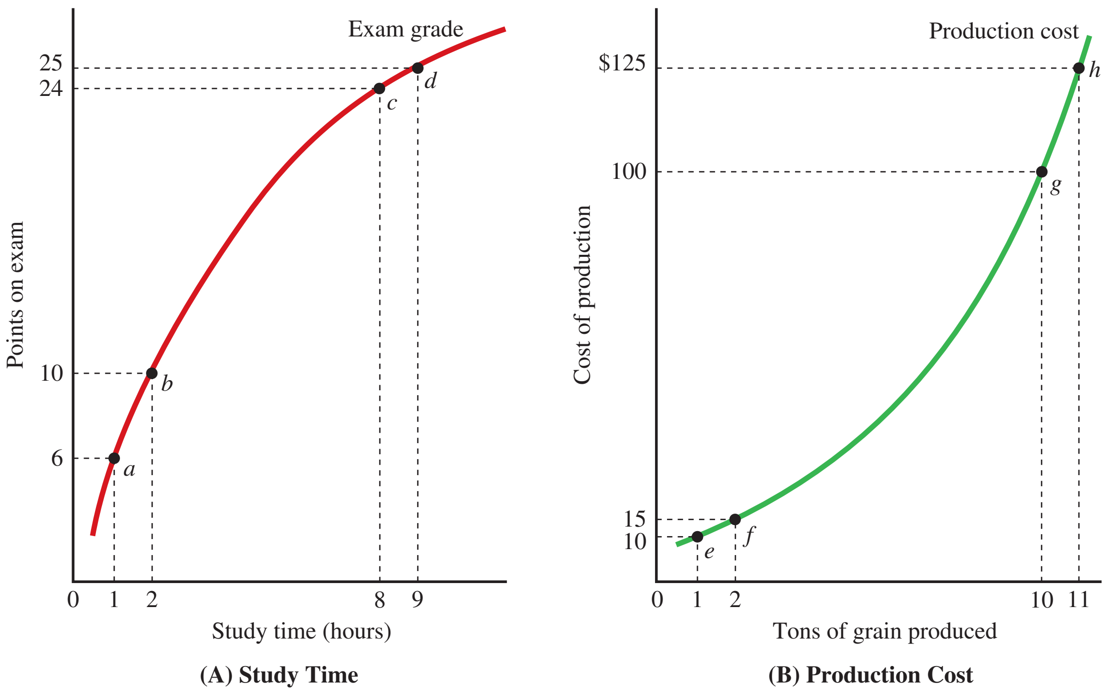
$$\text{Slope} = \frac{\text{Vertical difference}}{\text{Horizontal difference}} = \frac{120 - 240}{20 - 10} = \frac{-120}{10} = -12$$

The slope is 12 downloaded songs per CD: For each additional CD, the consumer sacrifices 12 downloaded songs. MyLab Economics Concept Check

Graphing Nonlinear Relationships

We can also use a graph to show a nonlinear relationship between two variables. Panel A of Figure 1A.7 shows the relationship between hours spent studying for an exam and the grade on the exam. As study time increases, the grade increases, but at a decreasing rate. In other words, each additional hour increases the exam grade by a smaller and smaller amount. For example, the second hour of study increases the grade by 4 points—from 6 to 10 points—but the ninth hour of study increases the grade by only 1 point—from 24 points to 25 points. This is a nonlinear relationship: The slope of the curve changes as we move along the curve. In Figure 1A.7, the slope decreases as we move to the right along the curve: The slope is 4 points per hour between points *a* and *b* but only 1 point per hour between points *c* and *d*.

Another possibility for a nonlinear curve is that the slope increases as we move to the right along the curve. Panel B of Figure 1A.7 shows the relationship between



▲ **FIGURE 1A.7** Nonlinear Relationships

(A) Study time There is a positive and nonlinear relationship between study time and the grade on an exam. As study time increases, the exam grade increases at a decreasing rate. For example, the second hour of study increased the grade by 4 points (from 6 points to 10 points), but the ninth hour of study increases the grade by only 1 point (from 24 points to 25 points).

(B) Production cost There is a positive and nonlinear relationship between the quantity of grain produced and total production cost. As the quantity increases, the total cost increases at an increasing rate. For example, to increase production from 1 ton to 2 tons, production cost increases by $5 (from $10 to $15) but to increase the production from 10 to 11 tons, total cost increases by $25 (from $100 to $125).

the amount of grain produced on the horizontal axis and the total cost of production on the vertical axis. The slope of the curve increases as the amount of grain increases, meaning that production cost increases at an increasing rate. On the lower part of the curve, increasing output from 1 ton to 2 tons increases production cost by $5, from $10 to $15. On the upper part of the curve, increasing output from 10 to 11 tons increases production cost by $25, from $100 to $125. MyLab Economics Concept Check

MyLab Economics Study Plan

1A.2 Computing Percentage Changes and Using Equations

Economists often express changes in variables in terms of percentage changes. This part of the appendix provides a brief review of the mechanics of computing percentage changes. It also reviews some simple rules for solving equations to find missing values.

Computing Percentage Changes

In many cases, the equations that economists use involve percentage changes. In this text, we use a simple approach to computing percentage changes: We divide the change in the variable by the initial value of the variable and then multiply by 100:

$$\text{Percentage change} = \frac{\text{New value} - \text{initial value}}{\text{Initial value}} \times 100$$

For example, if the price of a book increases from $20 to $22, the percentage change is 10 percent:

$$\text{Percentage change} = \frac{22 - 20}{20} \times 100 = \frac{2}{20} \times 100 = 10\%$$

Going in the other direction, suppose the price decreases from $20 to $19. In this case, the percentage change is −5 percent:

$$\text{Percentage change} = \frac{19 - 20}{20} \times 100 = -\frac{1}{20} \times 100 = -5\%$$

The alternative to this simple approach is to base the percentage change on the average value, or the midpoint, of the variable:

$$\text{Percentage change} = \frac{\text{New value} - \text{initial value}}{\text{Average value}} \times 100$$

For example, if the price of a book increases from $20 to $22, the computed percentage change under the midpoint approach is 9.52 percent:

$$\text{Percentage change} = \frac{22 - 20}{(20 + 22) \div 2} \times 100 = \frac{2}{42 \div 2} \times 100$$

$$= \frac{2}{21} \times 100 = 9.52\%,$$

If the change in the variable is relatively small, the extra precision associated with the midpoint approach is usually not worth the extra effort. The simple approach allows us to spend less time doing tedious arithmetic and more time doing economic analysis. In this text, we use the simple approach to compute percentage changes: If the price increases from $20 to $22, the price has increased by 10 percent.

If we know a percentage change, we can translate it into an absolute change. For example, if a price has increased by 10 percent and the initial price is $20, then we add 10 percent of the initial price ($2 is 10 percent of $20) to the initial price ($20), for a new price of $22. If the price decreases by 5 percent, we subtract 5 percent of the initial price ($1 is 5 percent of $20) from the initial price ($20), for a new price of $19. MyLab Economics Concept Check

Application 3

THE PERILS OF PERCENTAGES

APPLYING THE CONCEPTS #3: How do we compute percentage changes?

In the 1970s the government of Mexico City repainted the highway lane lines on the *Viaducto* to transform a four-lane highway into a six-lane highway. The government announced that the highway capacity had increased by 50 percent (equal to 2 divided by 4). Unfortunately, the number of collisions and traffic fatalities increased, and one year later, the government restored the four-lane highway and announced that the capacity had decreased by 33 percent (equal to 2 divided by 6). The government announced that the net effect of the two changes was an increase in the highway capacity by 17 percent (equal to 50 percent minus 33 percent).

This anecdote reveals a potential problem with using the simple approach to compute percentage changes. Because the initial value (the denominator) changes, the computation of percentage increases and decreases are not symmetric. In contrast, if the government had used the midpoint method, the percentage increase in capacity would be 40 percent (equal to 2 divided by 5), the same as the percentage decrease. In that case, we get the more sensible result that the net effect of the two changes is zero. **Related to Exercise A8.**

SOURCE: Based on "The Perils of Percentages," The Economist, April 18, 1998, 70.

Using Equations to Compute Missing Values

It will often be useful to compute the value of the numerator or the denominator of an equation. To do so, we use simple algebra to rearrange the equation to put the missing variable on the left side of the equation. For example, consider the relationship between time worked and income. The equation for the slope is

$$\text{Slope} = \frac{\Delta \text{ Income}}{\Delta \text{ Work hours}}$$

Suppose you want to compute how much income you'll earn by working more hours. We can rearrange the slope equation by multiplying both sides of the equation by the change in work hours:

$$\text{Work hours} \times \text{Slope} = \Delta \text{ Income}$$

By swapping sides of the equation, we get

$$\Delta \text{ Income} = \Delta \text{ Work hours} \times \text{Slope}$$

For example, if you work seven extra hours and the slope is \$8, your income will increase by \$56:

$$\Delta \text{ Income} = \Delta \text{ Work hours} \times \text{Slope} = \$7 \times \$8 = \$56$$

We can use the same process to compute the difference in work time required to achieve a target change in income. In this case, we multiply both sides of the slope equation by the change in work time and then divide both sides by the slope. The result is

$$\Delta \text{ Work hours} = \frac{\Delta \text{ Income}}{\text{Slope}}$$

For example, to increase your income by \$56, you need to work seven hours:

MyLab Economics Study Plan

$$\Delta \text{ Work hours} = \frac{\Delta \text{ Income}}{\text{Slope}} = \frac{\$56}{\$8} = 7$$

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KEY TERMS

negative relationship, p. 49 **positive relationship**, p. 49 **slope of a curve**, p. 49

EXERCISES

All problems are assignable in MyLab Economics.

A.1 Suppose you belong to a tennis club that has a monthly fee of \$100 and a charge of \$5 per hour to play tennis.

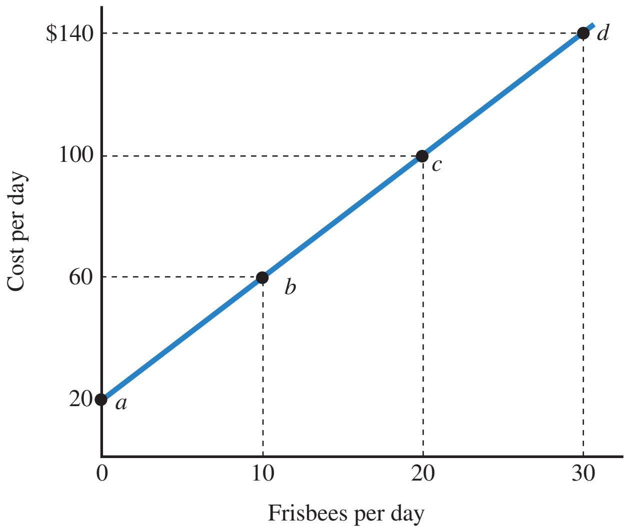
a. Using Figure 1A.4 on page 48 as a model, prepare a table and draw a curve to show the relationship between the hours of tennis (on the horizontal axis) and the monthly club bill (on the vertical axis). For the table and graph, use 5, 10, 15, and 20 hours of tennis.

b. The slope of the curve is _________ per _________.

c. Suppose you start with 10 hours of tennis and then decide to increase your tennis time by 3 hours. On your curve, show the initial point and the new point. By how much will your monthly bill increase?

d. Suppose you start with 10 hours and then decide to spend an additional \$30 on tennis. On your curve, show the initial point and the new point. How many additional hours can you get?

A.2 The following graph shows the relationship between the number of Frisbees produced and the cost of production. The vertical intercept is \$_________, and the slope of the curve is \$_________ per Frisbee. Point *b* shows that the cost of producing _________ Frisbees is \$_________. The cost of producing 15 Frisbees is \$_________.



A.3 Suppose you have \$120 to spend on CDs and movies. The price of a CD is \$12, and the price of a movie is \$6.

a. Using Figure 1A.6 on page 51 as a model, prepare a table and draw a curve to show the relationship between the number of CDs (on the horizontal axis) and movies (on the vertical axis) you can afford to buy.

b. The slope of the curve is _________ per _________.

A.4 You manage Gofer Delivery Service. You rent a truck for \$50 per day, and each delivery takes an hour of labor time. The hourly wage is \$8.

a. Draw a curve showing the relationship between the number of deliveries (on the horizontal axis) and your total cost (on the vertical axis). Draw the curve for between 0 and 20 deliveries.

b. The slope of the cost curve is _________ per _________.

c. To draw the curve, what variables are held fixed?

d. A change in _________ would cause a movement upward along the curve.

e. Changes in _________ would cause the entire curve to shift upward.

A.5 A change in a variable measured on an axis of a graph causes movement _________ a curve, while a change in a relevant variable that is not measured on an axis _________ the entire curve.

A.6 Compute the percentage changes for the following:

Initial Value	New Value	Percentage Change
10	12	
100	92	
40	45	

A.7 Compute the new values for the following changes:

Initial Value	Percentage Change	New Value
100	12%	
50	8	
20	15	

A.8 Suppose the price of an MP3 player decreases from \$60 to \$40. Using the midpoint approach, the percentage change in price is _________. Using the initial-value approach, the percentage change in price is _________. (Related to Application 3 on page 54.)

NOTES

1. Texas Transportation Institute, *2005 Urban Mobility Study*, http://mobility.tamu.edu/ums/.

2. William Easterly, *The Elusive Quest for Growth* (Cambridge, MA: MIT Press, 2001), Chapter 1.

3. William Easterly, *The Elusive Quest for Growth* (Cambridge, MA: MIT Press, 2001); World Bank, *World Development Report 2000/2001: Attacking Poverty* (New York: Oxford University Press, 2000).

4. John Maynard Keynes, *The Collected Writings of John Maynard Keynes*, *Volume* 7, ed. Donald Moggridge (London: Macmillan, 1973), 856. Reproduced with permission of the publisher.

5. Alfred Marshall, *Principles of Economics*, 9th ed., ed. C.W. Guillebaud (1920; repr., London: Macmillan, 1961), 366. Reproduced with permission of the publisher.

6. Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Book 2, Chapter 3.

CHAPTER

2

The Key Principles of Economics

What do we sacrifice by preserving tropical rainforests rather than mining or logging the land?

Recent experiences in Guyana and other tropical countries suggest that in some places, the answer is "not much"—only $1 per hectare per year ($0.40 per acre per year).[1] Conservation groups have a new strategy for conserving rainforests—bidding against loggers and miners for the use of the land. When the payoff from developing tropical forest land is relatively low, conservation groups can outbid developers at a price as low as $1 per hectare. When we add the cost of hiring locals to manage the ecosystems, the total cost of preservation is as low as $2 per hectare per year. A conservation group based in Amherst, New Hampshire, started by leasing 81,000 hectares of pristine forest in Guyana, and since then has leased land in Peru, Sierra Leone, Papua New Guinea, Fiji, and Mexico.

CHAPTER OUTLINE AND LEARNING OBJECTIVES

MyLab Economics
MyLab Economics helps you master each objective and study more efficiently.

In this chapter, we introduce five key principles that provide a foundation for economic analysis. A *principle* is a self-evident truth that most people readily understand and accept. For example, most people readily accept the principle of gravity. As you read through the book, you will see the five key principles of economics again and again as you do your own economic analysis.

Learning Objective 2.1

Apply the principle of opportunity cost.

The Principle of Opportunity Cost

opportunity cost
What you sacrifice to get something.

Economics is all about making choices, and to make good choices we must compare the benefit of something to its cost. **Opportunity cost** incorporates the notion of scarcity: No matter what we do, there is always a trade-off. We must trade off one thing for another because resources are limited and can be used in different ways. By acquiring something, we use up resources that could have been used to acquire something else. The notion of opportunity cost allows us to measure this trade-off.

PRINCIPLE OF OPPORTUNITY COST

The opportunity cost of something is what you sacrifice to get it.

In most decisions we choose from several alternatives. For example, if you spend an hour studying for an economics exam, you have one fewer hour to pursue other activities. To determine the opportunity cost of an activity, we look at what you consider the best of these "other" activities. For example, suppose the alternatives to studying economics are studying for a history exam or working in a job that pays $10 per hour. If you consider studying for history a better use of your time than working, then the opportunity cost of studying economics is the four extra points you could have received on a history exam if you studied history instead of economics. Alternatively, if working is the best alternative, the opportunity cost of studying economics is the $10 you could have earned instead.

We can also apply the principle of opportunity cost to decisions about how to spend money from a fixed budget. For example, suppose that you have a fixed budget to spend on music. You can buy your music either at a local music store for $15 per CD or online for $1 per song. The opportunity cost of 1 CD is 15 one-dollar online songs. A hospital with a fixed salary budget can increase the number of doctors only at the expense of nurses or physician's assistants. If a doctor costs five times as much as a nurse, the opportunity cost of a doctor is five nurses.

In some cases, a product that appears to be free actually has a cost. That's why economists are fond of saying, "There's no such thing as a free lunch." Suppose someone offers to buy you lunch if you agree to listen to a sales pitch for a timeshare condominium. Although you don't pay any money for the lunch, there is an opportunity cost because you could spend that time in another way—such as studying for your economics or history exam. The lunch isn't free because you sacrifice an hour of your time to get it.

The Cost of College

What is the opportunity cost of a college degree? Consider a student who spends a total of $40,000 for tuition and books. Instead of going to college, the student could have spent this money on a wide variety of goods, including housing, electronic devices, and world travel. Part of the opportunity cost of college is the $40,000 worth of other goods the student sacrifices to pay for tuition and books. Also, instead of going to college, the student could have worked as a bank clerk for $20,000 per year and earned $80,000 over four years. That makes the total opportunity cost of this student's college degree $120,000:

Opportunity cost of money spent on tuition and books	$ 40,000
Opportunity cost of college time (four years working for $20,000 per year)	80,000
Economic cost or total opportunity cost	$120,000

We haven't included the costs of food or housing in our computations of opportunity cost. That's because a student must eat and live somewhere even if he or she doesn't go to college. But if housing and food are more expensive in college, then we would include the extra costs of housing and food in our calculations.

There are other things to consider in a person's decision to attend college. As we'll see later, a college degree can increase a person's earning power, so there are benefits from a college degree. In addition, college offers the thrill of learning and the pleasure of meeting new people. To make an informed decision about whether to attend college, we must compare the benefits to the opportunity costs. MyLab Economics Concept Check

The Cost of Military Spending

We can use the principle of opportunity cost to explore the cost of military spending.[2] In 1992, Malaysia bought two warships. For the price of the warships, the country could have built a system to provide safe drinking water for 5 million citizens who lacked it. In other words, the opportunity cost of the warships was safe drinking water for 5 million people. The policy question is whether the benefits of the warships exceed their opportunity cost.

In the United States, economists estimated that the cost of the war in Iraq will be at least $1 trillion. The economists' calculations go beyond the simple budgetary costs and quantify the opportunity cost of the war. For example, the resources used in the war could have been used in various government programs for children—to enroll more children in preschool programs, to hire more science and math teachers to reduce class sizes, or to immunize more children in poor countries. For example, each $100 billion spent on the war could instead support one of the following programs:

- Enroll 13 million preschool children in the Head Start program for one year.
- Hire 1.8 million additional teachers for one year.
- Immunize all the children in less-developed countries for the next 33 years.

The fact that a war has a large opportunity cost does not necessarily mean that it is unwise. A policy question is whether the benefits from a war exceed its opportunity cost. Taking another perspective, we can measure the opportunity cost of war in terms of its implications for domestic security. The resources used in the war in Iraq could have been used to improve domestic security by securing ports and cargo facilities, hiring more police officers, improving the screening of airline passengers and baggage, improving fire departments and other first responders, upgrading the Coast Guard fleet, and securing our railroad and highway systems. The cost of implementing the domestic-security recommendations of various government commissions would be about $31 billion, a small fraction of the cost of the war. The question for policymakers is whether money spent on domestic security would be more beneficial than money spent on the war. MyLab Economics Concept Check

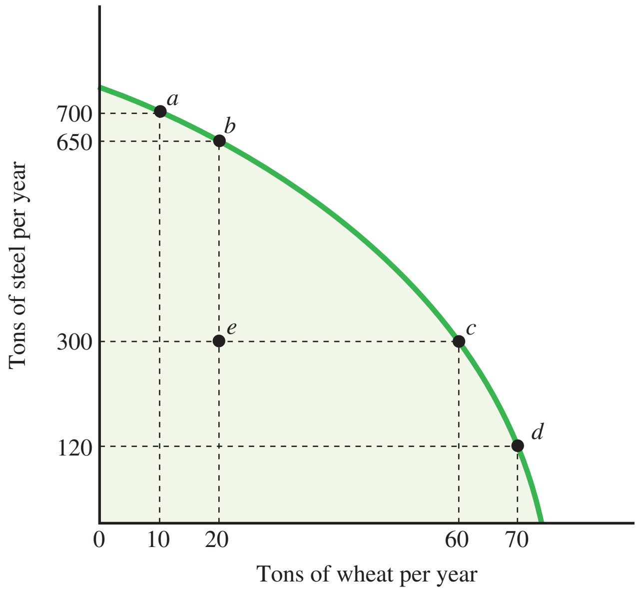
Opportunity Cost and the Production Possibilities Curve

Just as individuals face limits, so do entire economies. As we saw in Chapter 1, the ability of an economy to produce goods and services is determined by its factors of production, including labor, natural resources, physical capital, human capital, and entrepreneurship.

Figure 2.1 shows a production possibilities graph for an economy that produces wheat and steel. The horizontal axis shows the quantity of wheat produced by the economy, and the vertical axis shows the quantity of steel produced. The shaded area shows all the possible combinations of the two goods the economy can produce. At point *a*, for example, the economy can produce 700 tons of steel and 10 tons of wheat. In contrast, at point *e*, the economy can produce 300 tons of steel and 20 tons of wheat. The set of points on the border between the shaded and unshaded area is called the **production possibilities curve** (or *production possibilities frontier*) because it separates the combinations that are attainable from those that are not. The attainable combinations are shown by the shaded area within the curve and the curve itself. The unattainable combinations are shown by the unshaded area outside the curve. The points on the curve show the combinations that are possible if the economy's resources are fully employed.

production possibilities curve

A curve that shows the possible combinations of products that an economy can produce, given that its productive resources are fully employed and efficiently used.



▲ **FIGURE 2.1** Scarcity and the Production Possibilities Curve

The production possibilities curve illustrates the principle of opportunity cost for an entire economy. An economy has a fixed amount of resources. If these resources are fully employed, an increase in the production of wheat comes at the expense of steel.

The production possibilities curve illustrates the notion of opportunity cost. If an economy is fully utilizing its resources, it can produce more of one product only if it produces less of another product. For example, to produce more wheat, we must take resources away from steel. As we move resources out of steel, the quantity of steel produced will decrease. For example, if we move from point *a* to point *b* along the production possibilities curve in Figure 2.1, we sacrifice 50 tons of steel (700 tons − 650 tons) to get 10 more tons of wheat (20 tons − 10 tons). Further down the curve, if we move from point *c* to point *d*, we sacrifice 180 tons of steel to get the same 10-ton increase in wheat.

Why is the production possibilities curve bowed outward, with the opportunity cost of wheat increasing as we move down the curve? The reason is that resources are not perfectly adaptable for the production of both goods. Some resources are more suitable for steel production, while others are more suitable for wheat production. Starting at point *a*, the economy uses its most fertile land to produce wheat. A 10-ton increase in wheat reduces the quantity of steel by only 50 tons because plenty of fertile land is available for conversion to wheat farming. As the economy moves downward along the production possibilities curve, farmers will be forced to use land that is progressively less fertile, so to increase wheat output by 10 tons, more and more resources must be diverted from steel production. In the move from point *c* to point *d*, the land converted to farming is so poor that increasing wheat output by 10 tons decreases steel output by 180 tons.The production possibilities curve shows the production options for a given set of resources. As shown in Figure 2.2, an increase in the amount of resources available to the economy shifts the production possibilities outward. For example, if we start at point *f*, and the economy's resources increase, we can produce more steel (point *g*), more wheat (point *h*), or more of both goods (points between *g* and *h*). The curve will also shift outward as a result of technological innovations that enable us to produce more output with a given quantity of resources. MyLab Economics Concept Check

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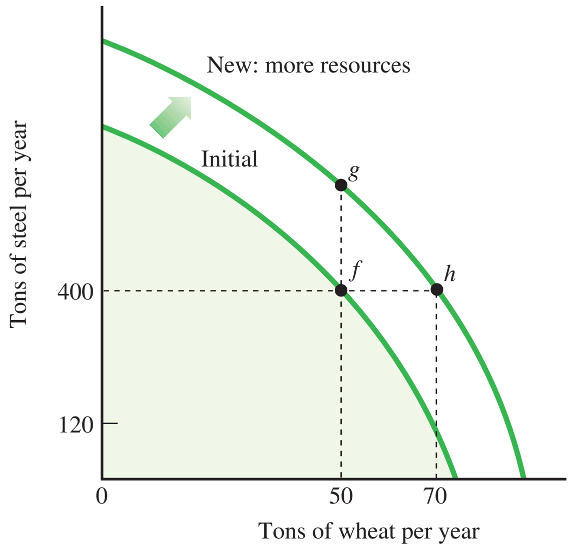


▲ **FIGURE 2.2** Shifting the Production Possibilities Curve

An increase in the quantity of resources or technological innovation in an economy shifts the production possibilities curve outward. Starting from point *f*, a nation could produce more steel (point *g*), more wheat (point *h*), or more of both goods (points between *g* and *h*).

Application 1

DON'T FORGET THE COSTS OF TIME AND INVESTED FUNDS

APPLYING THE CONCEPTS #1: What is the opportunity cost of running a business?

Suppose you run a lawn-cutting business and use solar-powered equipment (mower, edger, blower, truck) that you could sell tomorrow for $5,000. Instead of cutting lawns, you could work as a janitor for $300 per week. You have a savings account that pays a weekly interest rate of 0.20 percent (or $0.002 per dollar). What is your weekly cost of cutting lawns?

We can use the principle of opportunity cost to compute the cost of the lawn business. The opportunity cost of the $5,000 you have invested in the business is the $10 weekly interest you could have earned by selling the equipment and investing the $5,000 in your savings account. Adding in the opportunity cost of cutting lawns instead of earning $300 as a janitor, the weekly cost is $310. **Related to Exercise 1.7.**

Learning Objective 2.2

Apply the marginal principle.

The Marginal Principle

Economics is about making choices, and we rarely make all-or-nothing choices. For example, if you sit down to read a book, you don't read the entire book in a single sitting, but instead decide how many pages or chapters to read. Economists think in marginal terms, considering how a one-unit change in one variable affects the value of another variable and people's decisions. When we say *marginal*, we're looking at the effect of a small, or incremental, change.

marginal benefit
The additional benefit resulting from a small increase in some activity.

marginal cost
The additional cost resulting from a small increase in some activity.

The marginal principle is based on a comparison of the marginal benefits and marginal costs of a particular activity. The **marginal benefit** of an activity is the additional benefit resulting from a small increase in the activity. For example, the marginal benefit of keeping a bookstore open for one more hour equals the additional revenue from book sales. Similarly, the **marginal cost** is the additional cost resulting from a small increase in the activity. For example, the marginal cost of keeping a bookstore open for one more hour equals the additional expenses for workers and utilities for that hour. Applying the marginal principle, the bookstore should stay open for one more hour if the marginal benefit (the additional revenue) is at least as large as the marginal cost (the additional cost). For example, if the marginal benefit is $80 of additional revenue and the marginal cost is $30 of additional expense for workers and utilities, staying open for the additional hour increases the bookstore's profit by $50.

MARGINAL PRINCIPLE

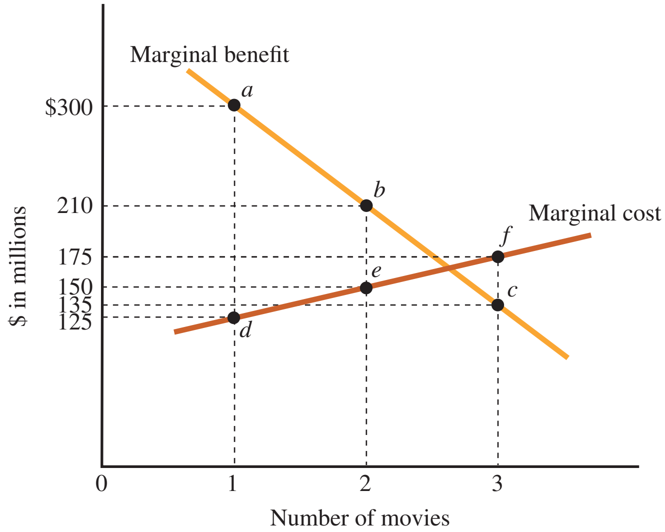
Increase the level of an activity as long as its marginal benefit exceeds its marginal cost. Choose the level at which the marginal benefit equals the marginal cost.

Thinking at the margin enables us to fine-tune our decisions. We can use the marginal principle to determine whether a one-unit increase in a variable would make us better off. Just as a bookstore owner could decide whether to stay open for one more hour, you could decide whether to study one more hour for a psychology midterm. When we reach the level where the marginal benefit equals the marginal cost, we cannot do any better, and the fine-tuning is done.

How Many Movie Sequels?

To illustrate the marginal principle, let's consider movie sequels. When a movie is successful, its producer naturally thinks about doing another movie, continuing the story line with the same set of characters. If the first sequel is successful too, the producer thinks about producing a second sequel, then a third, and so on. We can use the marginal principle to explore the decision of how many movies to produce.

Figure 2.3 shows the marginal benefits and marginal costs for movies. On the benefit side, a movie sequel typically generates about 30 percent less revenue than the original movie, and revenue continues to drop for additional movies. In the second column of the table, the first movie generates $300 million in revenue, the second generates $210 million, and the third generates $135 million. This is shown in the graph as a negatively sloped marginal-benefit curve, with the marginal benefit decreasing from $300 for the first movie (point *a*), to $210 (point *b*), and then to $135 (point *c*). On the cost side, the typical movie in the United States costs about $50 million to produce and about $75 million to promote.[3] In the third column of the table, the cost of the first movie (the original) is $125 million. In the graph, this is shown as point *d* on the marginal-cost curve. The marginal cost increases with the number of movies because film stars typically demand higher salaries to appear in sequels. In the table and the graph, the marginal cost increases to $150 million for the second movie (point *e*) and to $175 million for the third (point *f*).



▲ **FIGURE 2.3** The Marginal Principle and Movie Sequels

The marginal benefit of movies in a series decreases because revenue falls off with each additional movie, while the marginal cost increases because actors demand higher salaries. The marginal benefit exceeds the marginal cost for the first two movies, so it is sensible to produce two, but not three, movies.

Number of Movies	Marginal Benefit ($ millions)	Marginal Cost ($ millions)
1	$300	$125
2	210	150
3	135	175

In this example, the first two movies are profitable, but the third is not. For the original movie, the marginal benefit ($300 million at point *a*) exceeds the marginal cost ($125 million at point *d*), generating a profit of $175 million. Although the second movie has a higher cost and a lower benefit, it is profitable because the marginal benefit still exceeds the marginal cost, so the profit on the second movie is $60 million ($210 million − $150 million). In contrast, the marginal cost of the third movie of $175 million exceeds its marginal benefit of only $135 million, so the third movie *loses* $40 million. In this example, the movie producer should stop after the second movie.

Although this example shows that only two movies are profitable, other outcomes are possible. If the revenue for the third movie were larger, making the marginal benefit greater than the marginal cost, it would be sensible to produce the third movie. Similarly, if the marginal cost of the third movie were lower—if the actors didn't demand such high salaries—the third movie could be profitable. Many movies have had multiple sequels, such as *Harry Potter* and *Star Wars*. Conversely, many profitable movies, such as *Wedding Crashers* and *Groundhog Day*, didn't result in any sequels. In these cases, the expected drop-off in revenues and run-up in costs for the second movie were large enough to make a sequel unprofitable. MyLab Economics Concept Check

Renting College Facilities

Suppose that your student film society is looking for an auditorium to use for an all-day Hitchcock film program and is willing to pay up to $200. Your college has a new auditorium that has a daily rent of $450, an amount that includes $300 to help pay for the cost of building the auditorium, $50 to help pay for insurance, and $100 to cover the extra costs of electricity and janitorial services for a one-day event. If your film society offers to pay $150 for using the auditorium, should the college accept the offer? The college could use the marginal principle to make the decision.

To decide whether to accept your group's offer, the college should determine the marginal cost of renting out the auditorium. The marginal cost equals the extra costs the college incurs by allowing the student group to use an otherwise vacant auditorium. In our example, the extra cost is $100 for additional electricity and janitorial services. It would be sensible for the college to rent the auditorium because the marginal benefit ($150 offered by the student group) exceeds the marginal cost ($100). In fact, the college should be willing to rent the facility for any amount greater than $100. If the students and the college split the difference between the $200 the students are willing to pay and the $100 marginal cost, they would agree on a price of $150, leaving both parties better off by $50.

Most colleges do not use this sort of logic. Instead, they use complex formulas to compute the perceived cost of renting out a facility. In most cases, the perceived cost includes some costs that the university bears even if it doesn't rent out the facility for the day. In our example, the facility manager included $300 worth of construction costs and $50 worth of insurance, for a total cost of $450 instead of just $100. Because many colleges include costs that aren't affected by the use of a facility, they overestimate the actual cost of renting out their facilities, missing opportunities to serve student groups and make some money at the same time. MyLab Economics Concept Check

Automobile Emissions Standards

We can use the marginal principle to analyze emissions standards for automobiles. The U.S. government specifies how much carbon monoxide a new car is allowed to emit per mile. The marginal question is "Should the standard be stricter, with fewer units of carbon monoxide allowed?" On the benefit side, a stricter standard reduces health-care costs resulting from pollution: If the air is cleaner, people with respiratory ailments will make fewer visits to doctors and hospitals, have lower medication costs, and lose fewer work days. On the cost side, a stricter standard requires more expensive control equipment on cars and may also reduce fuel efficiency. Using the marginal principle, the government should make the emissions standard stricter as long as the marginal benefit (savings in health-care costs and work time lost) exceeds the marginal cost (the cost of additional equipment and extra fuel used). MyLab Economics Concept Check

Driving Speed and Safety

Consider the decision about how fast to drive on a highway. The marginal benefit of going one mile per hour faster is the travel time you'll save. On the cost side, an increase in speed increases your chances of colliding with another car, and also increases the severity of injuries suffered in a collision. A rational person will pick the speed at which the marginal benefit of speed equals the marginal cost.

In the 1960s and 1970s, the federal government required automakers to include a number of safety features, including seat belts and collapsible steering columns. These new regulations had two puzzling effects. Although deaths from automobile collisions decreased, the reduction was much lower than expected. In addition, more bicyclists were hit by cars and injured or killed.

We can use the marginal principle to explain why seat belts and other safety features made bicycling more hazardous. The mandated safety features decreased the marginal cost of speed: People who wear seat belts suffer less severe injuries in a collision, so every additional unit of speed is less costly. Drivers felt more secure because they were better insulated from harm in the event of a collision, and so they drove faster. As a result, the number of collisions between cars and bicycles increased, meaning that the safer environment for drivers led to a more hazardous environment for bicyclists. MyLab Economics Concept Check

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Application 2

HOW FAST TO SAIL?

APPLYING THE CONCEPTS #2: How do people think at the margin?

Consider the decision about how fast to sail an ocean cargo ship. As the ship's speed increases, fuel consumption increases. For example, a 70,000-ton cargo ship burns 16.5 tons of fuel per day at 11 knots, compared to 21.4 tons at 12 knots, 27.2 tons at 13 knots, and 33.9 tons at 14 knots. In other words, speed is costly. On the benefit side, an increase in speed means that the ship delivers more cargo per year. To decide the best speed, the ship operator must find the speed at which the marginal cost (the increase in fuel cost) equals the marginal benefit (the increase in revenue from delivered cargo). An increase in the price of fuel increases the marginal cost of speed, causing the shipper to slow the ship. **Related to Exercise 2.4.**

SOURCE: Based on Martin Stopard, *Maritime Economics*, 3rd edition (New York: Routledge, 2007).

The Principle of Voluntary Exchange

Learning Objective 2.3
Apply the principle of voluntary exchange.

The principle of voluntary exchange is based on the notion that people act in their own self-interest. Self-interested people won't exchange one thing for another unless the trade makes them better off.

PRINCIPLE OF VOLUNTARY EXCHANGE

A voluntary exchange between two people makes both people better off.

Here are some examples.

- If you voluntarily exchange money for a college education, you must expect you'll be better off with a college education. The college voluntarily provides an education in exchange for your money, so the college must be better off, too.
- If you have a job, you voluntarily exchange your time for money, and your employer exchanges money for your labor services. Both you and your employer are better off as a result.

Exchange and Markets

Adam Smith stressed the importance of voluntary exchange as a distinctly human trait. He noticed[4]

> a propensity in human nature . . . to truck, barter, and exchange one thing for another. . . . It is common to all men, and to be found in no other . . . animals. . . . Nobody ever saw a dog make a fair and deliberate exchange of one bone for another with another dog.

As we saw in Chapter 1, a market is an institution or arrangement that enables people to exchange goods and services. If participation in a market is voluntary and people are well informed, both people in a transaction—buyer and seller—will be better off. The next time you see a market transaction, listen to what people say after money changes hands. If both people say "thank you," that's the principle of voluntary exchange in action: The double "thank you" reveals that both people are better off.

The alternative to exchange is *self-sufficiency*: Each of us could produce everything for him- or herself. As we'll see in the next chapter, it is more sensible to specialize, doing what we do best and then buying products from other people, who in turn are doing what they do best. For example, if you are good with numbers but an awful carpenter, you could specialize in accounting and buy furniture from Woody, who could specialize in making furniture and pay someone to do his bookkeeping. In general, exchange allows us to take advantage of differences in people's talents and skills. MyLab Economics Concept Check

Online Games and Market Exchange

As another illustration of the power of exchange, consider the virtual world of online games. Role-playing games such as *World of Warcraft* and *EverQuest* allow thousands of people to interact online, moving their characters through a landscape of survival challenges. Each player constructs a character—called an *avatar*—by choosing some initial traits for it. The player then navigates the avatar through the game's challenges, where it acquires skills and accumulates assets, including clothing, weapons, armor, and even magic spells.

The curious part about these role-playing games is that players use real-life auction sites, including eBay and Yahoo! Auctions, to buy products normally acquired in the game.[5] Byron, who wants a piece of armor for his avatar (say, a Rubicite girdle), can use eBay to buy one for $50 from Selma. The two players then enter the online game, and Selma's avatar transfers the armor to Byron's avatar. It is even possible to buy another player's avatar, with all its skills and assets. Given the time required to acquire various objects such as Rubicite girdles in the game and the prices paid for them on eBay, the implicit wage earned by the typical online player auctioning them off is $3.42 per hour: That's how much the player could earn by first taking the time to acquire the assets in the game and then selling them on eBay. MyLab Economics Concept Check

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Application 3

RORY MCILROY AND WEED-WHACKING

APPLYING THE CONCEPTS #3: What is the rationale for specialization and exchange?

Should Rory McIlroy whack his own weeds? The swinging skills that make Rory McIlroy one of the world's best golfers also make him a skillful weed whacker. His large estate has a lot of weeds, and it would take the best gardener 20 hours to take care of all of them. With his powerful and precise swing, Rory could whack down all the weeds in just one hour. Since Rory is 20 times more productive than the best gardener, should he take care of his own weeds?

We can use the principle of voluntary exchange to explain why Rory should hire the less productive gardener. Suppose Rory earns $1,000 per hour playing golf—either playing in tournaments or giving lessons. For Rory, the opportunity cost of weed whacking is $1,000—the income he sacrifices by spending an hour cutting weeds rather than playing golf. If the gardener charges $10 per hour, Rory could hire him to take care of the weeds for only $200. By switching one hour of his time from weed-whacking to golf, Rory earns $1,000 and incurs a cost of only $200, so he is better off by $800. Rory McIlroy specializes in what he does best, and then buys goods and services from other people. **Related to Exercise 3.5.**

The Principle of Diminishing Returns

Learning Objective 2.4

Apply the principle of diminishing returns.

Xena has a small copy shop, with one copying machine and one worker. When the backlog of orders piled up, she decided to hire a second worker, expecting that doubling her workforce would double the output of her copy shop from 500 pages per hour to 1,000. Xena was surprised when output increased to only 800 pages per hour. If she had known about the principle of diminishing returns, she would not have been surprised.

PRINCIPLE OF DIMINISHING RETURNS

Suppose output is produced with two or more inputs, and we increase one input while holding the other input or inputs fixed. Beyond some point—called the *point of diminishing returns*—output will increase at a decreasing rate.

Xena added a worker (one input) while holding the number of copying machines (the other input) fixed. Because the two workers must share a single copying machine, each worker spent some time waiting for the machine to be available. As a result, adding the second worker increased the number of copies, but did not double the output. With a single worker and a single copying machine, Xena has already reached the point of diminishing returns: As she increases the number of workers, output increases, but at a decreasing rate. The first worker increases output by 500 pages (from 0 to 500), but the second worker increases output by only 300 pages (from 500 to 800).

This principle of diminishing returns is relevant when we try to produce more output in an existing production facility (a factory, a store, an office, or a farm) by increasing the number of workers sharing the facility. When we add a worker to the facility, each worker becomes less productive because he or she works with a smaller piece of the facility: More workers share the same machinery, equipment, and factory space. As we pack more and more workers into the factory, total output increases, but at a decreasing rate.

It's important to emphasize that diminishing returns occurs because one of the inputs to the production process is fixed. When a firm can vary all its inputs, including the size of the production facility, the principle of diminishing returns is not relevant. For example, if a firm doubled all its inputs, building a second factory and hiring a second workforce, we would expect the total output of the firm to at least double. The principle of diminishing returns does not apply when a firm is flexible in choosing all its inputs.

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Application 4

FERTILIZER AND CROP YIELDS

APPLYING THE CONCEPTS #4: Do farmers experience diminishing returns?

The notion of diminishing returns applies to all inputs to the production process. For example, one of the inputs in the production of corn is nitrogen fertilizer. Suppose a farmer has a fixed amount of land (an acre) and must decide how much fertilizer to apply. The first 50-pound bag of fertilizer will increase the crop yield by a relatively large amount, but the second bag is likely to increase the yield by a smaller amount, and the third bag is likely to have an even smaller effect. Because the farmer is changing just one of the inputs, the output will increase, but at a decreasing rate. Eventually, additional fertilizer will actually decrease output as the other nutrients in the soil are overwhelmed by the fertilizer.

Table 2.1 shows the relationship between the amount of fertilizer and the corn output. The first 50-pound bag of fertilizer increases the crop yield from 85 to 120 bushels per acre, a gain of 35 bushels. The next bag of fertilizer increases the yield by only 15 bushels (from 120 to 135), followed by a gain of 9 bushels (from 135 to 144) and then a gain of only 3 bushels (from 144 to 147). The farmer experienced diminishing returns because the other inputs to the production process are fixed. **Related to Exercises 4.5 and 4.6.**

TABLE 2.1 Fertilizer and Corn Yield

Bags of Nitrogen Fertilizer	Bushels of Corn per Acre
0	85
1	120
2	135
3	144
4	147

Learning Objective 2.5

Apply the real-nominal principle.

The Real-Nominal Principle

One of the key ideas in economics is that people are interested not just in the amount of money they have but also in how much their money will buy.

REAL-NOMINAL PRINCIPLE

What matters to people is the real value of money or income—its purchasing power—not its “face” value.

To illustrate this principle, suppose you work in your college bookstore to earn extra money for movies and snacks. If your take-home pay is $10 per hour, is this a high wage or a low wage? The answer depends on the prices of the goods you buy. If a movie costs $4 and a snack costs $1, with one hour of work you could afford to see two movies and buy two snacks. The wage may seem high enough for you. But if a movie costs $8 and a snack costs $2, an hour of work would buy only one movie and one snack, and the same $10 wage doesn’t seem so high. This is the real-nominal principle in action: What matters is not how many dollars you earn, but what those dollars will purchase.

Economists use special terms to express the ideas behind the real-nominal principle:

nominal value
The face value of an amount of money.

real value
The value of an amount of money in terms of what it can buy.

- The **nominal value** of an amount of money is simply its face value. For example, the nominal wage paid by the bookstore is $10 per hour.
- The **real value** of an amount of money is measured in terms of the quantity of goods the money can buy. For example, the real value of your bookstore wage would fall as the prices of movies and snacks increase, even though your nominal wage stayed the same.

The real-nominal principle can explain how people choose the amount of money to carry around with them. Suppose you typically withdraw $40 per week from an ATM to cover your normal expenses. If the prices of all the goods you purchase during the week double, you would have to withdraw $80 per week to make the same purchases. The amount of money people carry around depends on the prices of the goods and services they buy.

The Design of Public Programs

Government officials use the real-nominal principle when they design public programs. For example, Social Security payments are increased each year to ensure that the checks received by the elderly and other recipients will purchase the same amount of goods and services, even if prices have increased. The government also uses this principle when it publishes statistics about the economy. For example, its reports about changes in "real wages" in the economy over time take into account the prices of the goods workers purchase. Therefore, the real wage is stated in terms of its buying power, rather than its face value or nominal value. MyLab Economics Concept Check

The Value of the Minimum Wage

Between 1974 and 2015, the federal minimum wage increased from $2.00 to $7.25. Was the typical minimum-wage worker better or worse off in 2015? We can apply the real-nominal principle to see what's happened over time to the real value of the federal minimum wage.

As shown in the second row of Table 2.2, a full-time worker earning the minimum wage earned $80 in 1974 and $290 in 2015. The third row of Table 2.2 shows the cost of a standard basket of consumer goods, which includes a standard mix of housing, food, clothing, and transportation. In 1974, consumer prices were relatively low, and the cost of buying all the goods in the standard basket was only $47. Between 1974 and 2011, consumer prices increased, and the cost of this standard basket of goods increased to $236.

The last row in Table 2.2 shows the purchasing power of the minimum wage. In 1974, the $80 in weekly income could buy 1.70 standard baskets of goods. Between 1974 and 2015, the weekly income more than tripled, but the cost of the standard basket of goods more than quadrupled. As a result, the weekly income of $290 in 2011 could buy only 1.23 baskets of goods. Because prices increased faster than the nominal wage, the real value of the minimum wage actually decreased over this period. MyLab Economics Concept Check

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TABLE 2.2 The Real Value of the Minimum Wage, 1974–2011

	1974	2015
Minimum wage per hour	$ 2.00	$ 7.25
Weekly income from minimum wage	80	290
Cost of a standard basket of goods	47	236
Number of baskets per week	1.70	1.23

Application 5

REPAYING STUDENT LOANS

APPLYING THE CONCEPTS #5: How does inflation affect lenders and borrowers?

Suppose you finish college with $20,000 in student loans and start a job that pays a salary of $40,000 in the first year. In 10 years, you must repay your college loans. Which would you prefer: stable prices, rising prices, or falling prices?

We can use the real-nominal principle to compute the real cost of repaying your loans. The first row of Table 2.3 shows the cost of the loan when all prices in the economy are stable—including the price of labor, your salary. In this case, your nominal salary in 10 years is $40,000, and the real cost of repaying your loan is the half year of work you must do to earn the $20,000 you owe. However, if all prices double over the 10-year period, your nominal salary will double to $80,000, and, as shown in the second row of Table 2.3, it will take you only a quarter of a year to earn $20,000 to repay the loan. In other words, a general increase in prices lowers the real cost of your loan. In contrast, if all prices decrease and your annual salary drops to $20,000, it will take you a full year to earn the money to repay the loan. In general, people who owe money prefer inflation (a general rise in prices) to deflation (a general drop in prices). **Related to Exercises 5.6 and 5.9.**

TABLE 2.3 Effect of Inflation and Deflation on Loan Repayment

Change in Prices and Wages	Annual Salary	Years of Work to Repay $20,000 Loan
Stable	$40,000	1/2 year
Inflation: Salary doubles	80,000	1/4 year
Deflation: Salary cut in half	20,000	1 year

Chapter Summary and Problems

SUMMARY

This chapter covers five key principles of economics, the simple, self-evident truths that most people readily accept. If you understand these principles, you are ready to read the rest of the book, which will show you how to do your own economic analysis.

1. Principle of opportunity cost. The opportunity cost of something is what you sacrifice to get it.
2. Marginal principle. Increase the level of an activity as long as its marginal benefit exceeds its marginal cost. Choose the level at which the marginal benefit equals the marginal cost.
3. Principle of voluntary exchange. A voluntary exchange between two people makes both people better off.
4. Principle of diminishing returns. Suppose that output is produced with two or more inputs, and we increase one input while holding the other inputs fixed. Beyond some point—called the *point of diminishing returns*—output will increase at a decreasing rate.
5. Real-nominal principle. What matters to people is the real value of money or income—its purchasing power—not the face value of money or income.

KEY TERMS

marginal benefit, p. 62

marginal cost, p. 62

nominal value, p. 68

opportunity cost, p. 58

production possibilities curve, p. 60

real value, p. 68

EXERCISES

All problems are assignable in MyLab Economics.

The Principle of Opportunity Cost

Apply the principle of opportunity cost.

1.1 Consider Figure 2.1 on page 60. Between points *c* and *d*, the opportunity cost of __________ tons of wheat is __________ tons of steel.

1.2 Arrow up or down: An increase in the wage for high school graduates __________ the opportunity cost of college.

1.3 Arrow up or down: An increase in the market interest rate __________ the economic cost of holding a $500 collectible for a year.

1.4 You just inherited a house with a market value of $200,000 and do not expect the market value to change. Each year, you will pay $1,800 for utilities and $1,000 in taxes. You can earn 4 percent interest on money in a bank account. Your cost of living in the house for a year is $__________.

1.5 What is the cost of a pair of warships purchased by Malaysia?

1.6 Conservationists have a new strategy for preserving rainforests: __________ loggers and other developers for the land, paying as little as $ __________ per hectare per year.

1.7 **The Cost of a Toy Store Business.** Tom left a job paying $50,000 per year to start his own toy store in a building he owns. The market value of the building is $300,000. He pays $60,000 per year for toys and other supplies, and has a bank account that pays 3 percent interest. The annual economic cost of Tom's business is __________. (Related to Application 1 on page 61.)

1.8 **The Opportunity Cost of a Mission to Mars.** The United States has plans to spend billions of dollars on a mission to Mars. List some of the possible opportunity costs of the mission. What resources will be used to execute the mission, and what do we sacrifice by using these resources in a mission to Mars?

1.9 **Interest Rates and ATM Trips.** Carlos, who lives in a country where interest rates are very high, goes to an ATM every day to get $10 of spending money. Art, who lives in a country with relatively low interest rates, goes to the ATM once a month to get $300 of spending money. Why does Carlos use the ATM more frequently?

1.10 **Correct the Cost Statements.** Consider the following statements about cost. For each incorrect statement, provide a correct statement about the relevant cost.

a. One year ago, I loaned a friend $100, and she just paid me back the whole $100. The loan didn't cost me anything.

b. An oil refinery bought a million barrels of oil a month ago, when the price was only $75 per barrel, compared to $120 today. The cost of using a barrel of oil to produce gasoline is $75.

c. Our new football stadium was built on land donated to the university by a wealthy alum. The cost of the stadium equals the $50 million construction cost.

d. If a commuter rides a bus to work and the bus fare is $2, the cost of commuting by bus is $2.

1.11 **Production Possibilities Curve.** Consider a nation that produces baseball mitts and soccer balls. The following table shows the possible combinations of the two products.

Baseball mitts (millions)	0	2	4	6	8
Soccer balls (millions)	30	24	18	10	0

a. Draw a production possibilities curve with mitts on the horizontal axis and balls on the vertical axis.

b. Suppose the technology for producing mitts improves, meaning that fewer resources are needed for each mitt. In contrast, the technology for producing soccer balls does not change. Draw a new production possibilities curve.

c. The opportunity cost of the first 2 million mitts is __________ million soccer balls and the opportunity cost of the last 2 million mitts is __________ million soccer balls.

1.12 **Cost of Antique Furniture.** Colleen owns antique furniture that she bought for $5,000 ten years ago. Your job is to compute Colleen's cost of owning the furniture for the next year. To compute the cost, you need two bits of information: __________ and __________.

The Marginal Principle

Apply the marginal principle.

2.1 A car rental company currently has seven cars in its fleet, and its total daily cost is $2,300. If the company added an 8th car, its daily cost would be $2,400, or $300 per car. Adding the 8th car will increase the daily total revenue by $200. Should the company add the 8th car? __________ (Yes/No)

2.2 In Figure 2.3 on page 63, suppose the marginal cost of movies is constant at $125 million. Is it sensible to produce the third movie? __________ (Yes/No)

2.3 Suppose that stricter emissions standards would reduce health-care costs by $50 million but increase the costs of fuel and emissions equipment by $30 million. Is it sensible to tighten the emissions standards? __________ (Yes/No)

2.4 Arrows up or down: The decision about how fast to sail a cargo ship is based on the __________ benefit and the __________ cost. (Related to Application 2 on page 65.)

2.5 **How Fast to Drive?** Suppose Duke is driving to a nearby town to attend a dance party, and must decide how fast to drive. The marginal benefit of speed is the extra dance time he will get by driving faster, and the marginal cost is the additional risk of a collision. The marginal-benefit curve is negatively sloped, and the marginal-cost curve is positively sloped.

- **a.** Draw a pair of curves that suggest Duke will drive 40 mph.
- **b.** Suppose the normal country band is replaced by Adam Smith and the Invisible Hands, Duke's favorite punk band. Duke's utility from slam dancing is twice his utility from the two-step. Use your curves to show how Duke's chosen speed will change.
- **c.** Suppose Duke's favorite dance partner, Daisy, is grounded for makeup violations. Use your curves to show how Duke's chosen speed will change.
- **d.** Suppose the legal speed limit is set at 35 mph, and there is a 50 percent chance that Duke will be caught if he speeds. Use your curves to show how Duke's chosen speed will change.

2.6 **Highlanders Go Marginal.** Suppose that the owners of traditional horse-drawn carriages in the Tatra mountains of Poland observed that up to half the seats during evening rides through the Chochołowska Valley are empty. The average cost of an evening ride through the valley is 150 Polish zloty, a figure that includes fixed costs such as stable fees. A half-full evening ride generates only 200 Polish zloty of revenue.

- **a.** Use the marginal principle to explain why the owners of horse-drawn carriages continued evening rides through the valley.
- **b.** It will be sensible to run a half-empty evening ride if the marginal __________ of the ride is __________ than __________ Polish zloty.

2.7 **Marginal Airlines.** Marginal Airlines runs 10 flights per day at a total cost of $50,000, including $30,000 in fixed costs for airport fees and the reservation system and $20,000 for flight crews and food service.

- **a.** If an 11th flight would have 25 passengers, each paying $100, would it be sensible to run the flight?
- **b.** If the 11th flight would have only 15 passengers, would it be sensible to run the flight?

2.8 **How Many Police Officers?** In your city, each police officer has a budgetary cost of $40,000 per year. The property loss from each burglary is $4,000. The first officer hired will reduce crime by 40 burglaries, and each additional officer will reduce crime by half as much as the previous one. How many officers should the city hire? Illustrate with a graph with a marginal-benefit curve and a marginal-cost curve.

2.9 **How Many Hours at the Candy Stall?** The opportunity cost of your time spent selling candy at your stall in a shopping mall is $15 per hour. Electricity costs $2 per hour, and the weekly rent is $300. You normally stay open 11 hours per day (the mall is open 24/7).

- **a.** What is the marginal cost of staying open for one more hour?
- **b.** If you expect to sell 4 kilograms of candy in the 12th hour, earning $7 per kilogram, is it sensible to stay open for the extra hour?

2.10 **How Many Pints of Blackberries?** The pleasure you get from each pint of freshly picked blackberries is $2.00. It takes you 12 minutes to pick the first pint, and each additional pint takes an additional 2 minutes (14 minutes for the second pint, 16 minutes for the third pint, and so on). The opportunity cost of your time is $0.10 per minute.

- **a.** How many pints of blackberries should you pick? Illustrate with a complete graph.
- **b.** How would your answer to (a) change if your pleasure decreased by $0.20 for each additional pint ($1.80 for the second, $1.60 for the third, and so on)? Illustrate with a complete graph.

The Principle of Voluntary Exchange

Apply the principle of voluntary exchange.

3.1 When two people involved in an exchange say "thank you" afterward, they are merely being polite. __________ (True/False)

3.2 Consider a transaction in which a consumer buys a toy for $20. The value of the toy to the buyer is at least $__________, and the cost of producing the toy is no more than $__________.

3.3 Arrow up or down: Andy buys and eats one apple per day, and smacks his lips in appreciation as he eats it. The greater his satisfaction with the exchange of money for an apple, the larger the number of smacks. If the price of apples decreases, the number of smacks per apple will __________.

3.4 Sally sells one apple per day to Andy, and says "caching" to show her satisfaction with the transaction. The greater her satisfaction with the exchange, the louder her "caching." If the price of apples decreases, her "caching" will become __________ (louder/softer).

3.5 **Should a Heart Surgeon Do Her Own Plumbing?** A heart surgeon is skillful at unplugging arteries and rerouting the flow of blood, and these skills also make her a very skillful plumber. She can clear a clogged drain in six minutes, about 10 times faster than the most skillful plumber in town. (Related to Application 3 on page 66.)

a. Should the surgeon clear her own clogged drains? Explain.

b. Suppose the surgeon earns $20 per minute in heart surgery, and the best plumber in town charges $50 per hour. How much does the surgeon gain by hiring the plumber to clear a clogged drain?

3.6 **Fishing versus Boat-Building.** Half the members of a fishing tribe catch two fish per day and half catch eight fish per day. A group of 10 members could build a boat for another tribe in one day and receive a payment of 40 fish for the boat.

a. Suppose the boat builders are drawn at random from the tribe. From the tribe's perspective, what is the expected cost of building the boat?

b. How could the tribe decrease the cost of building the boat, thus making it worthwhile?

3.7 **Solving a Tree-Cutting Problem.** Consider a hilly neighborhood where large trees provide shade but also block views. When a resident announces plans to cut down several trees to improve her view, her neighbors object and announce plans to block the tree-cutting. One week later, the trees are gone, but everyone is happy. Use the principle of voluntary exchange to explain what happened.

The Principle of Diminishing Returns

Apply the principle of diminishing returns.

4.1 Consider the example of Xena's copy shop. Adding a second worker increased output by 300 pages. If she added a third worker, her output would increase by fewer than __________ pages.

4.2 If a firm is subject to diminishing marginal returns, an increase in the number of workers decreases the quantity produced. __________ (True/False)

4.3 Fill in the blanks with "at least" or "less than": If a firm doubles one input but holds the other inputs fixed, we normally expect output to __________ double; if a firm doubles all inputs, we expect output to __________ double.

4.4 Fill in the blanks with "flexible" or "inflexible": Diminishing returns is applicable when a firm is __________ in choosing inputs, but does not apply when a firm is __________ in choosing its inputs.

4.5 Arrows up or down: As a farmer adds more and more fertilizer to the soil, the crop yield __________, but at a __________ rate. (Related to Application 4 on page 68.)

4.6 **Feeding the World from a Flowerpot?** Comment on the following statement: "If agriculture did not experience diminishing returns, we could feed the world using the soil from a small flowerpot." (Related to Application 4 on page 68.)

4.7 **When to Use the Principle of Diminishing Returns?** You are the manager of a firm that produces memory chips for mobile phones.

a. In your decision about how much output to produce this week, would you use the principle of diminishing returns? Explain.

b. In your decision about how much output to produce two years from now, would you use the principle of diminishing returns? Explain.

4.8 **Diminishing Returns in a School?** Suppose you have two English teachers in your language school, and each of them works 6 hours a day. The school has one classroom.

a. If you double the number of English teachers but don't add a second classroom, would you expect the number of lessons offered by your school to double? Explain.

b. If you double the number of English teachers and add a second classroom, would you expect the number of lessons offered by your school to double? Explain.

4.9 **Diminishing Returns and the Marginal Principle.** Molly's Espresso Shop has become busy, and the more hours Ted works, the more espressos Molly can sell. The price of espressos is $2 and Ted's hourly wage is $11. Complete the following table:

Hours for Ted	Espressos Sold	Marginal Benefit from Additional Hour	Marginal Cost from Additional Hour
0	100	—	—
1	130	$60 = $2 × 30 additional espressos	$11 = hourly wage
2	154		
3	172		
4	184		
5	190		
6	193		

If Molly applies the marginal principle, how many hours should Ted work?

The Real-Nominal Principle

Apply the real-nominal principle.

5.1 Your savings account pays 4 percent per year: Each $100 in the bank grows to $104 over a 1-year period. If prices increase by 3 percent per year, by keeping $100 in the bank for a year you actually gain $__________.

5.2 You earn 4 percent interest on funds in your money-market account. If consumer prices increase by 5 percent a year, your earnings on $2,000 in the money-market account is $__________ per year.

5.3 Suppose that over a 1-year period, the nominal wage increases by 2 percent and consumer prices increase by 5 percent. Fill in the blanks: The real wage __________ by __________ percent.

5.4 Suppose you currently live and work in Cleveland, earning a salary of $60,000 per year and spending $10,000 for housing. You just heard that you will be transferred to a city in California where housing is 50 percent more expensive. In negotiating a new salary, your objective is to keep your real income constant. Your new target salary is $__________.

5.5 Between 1974 and 2011, the federal minimum wage increased from $2.00 to $7.25. Was the typical minimum-wage worker better off in 2011? __________ (Yes/No)

5.6 Suppose you graduate with $20,000 in student loans and repay the loans 10 years later. Which is better for you, inflation (rising prices) or deflation (falling prices)? __________ (Related to Application 5 on page 70.)

5.7 **Changes in Unemployment Benefits.** Between 2003 and 2013, the monthly unemployment benefit in Poland increased from 498 Polish zloty to 824 Polish zloty. Over the same period, the cost of a standard consumer basket of goods (a standard bundle of food, housing, and other goods and services) increased from 355 Polish zloty to 542 Polish zloty. Fill in the blanks in the following table. Did the real value of unemployment benefits increase or decrease over this period?

	2003	2013
Monthly unemployment benefits	PLN 498	PLN 824
Cost of a minimum consumer basket of goods	355	542
Number of baskets per week		

5.8 **Changes in Wages and Consumer Prices.** The following table shows for 1980 and 2004 the cost of a standard basket of consumer goods (a standard bundle of food, housing, and other goods and services) and the nominal average wage (hourly earnings) for workers in several sectors of the economy.

Year	Cost of Consumer Basket	Nominal Wage: Manufacturing	Nominal Wage: Professional Services	Nominal Wage: Leisure and Hospitality	Nominal Wage: Information
1980	$ 82	$ 7.52	$ 7.48	$4.05	$ 9.83
2004	189	16.34	17.69	9.01	21.70
Percent change from 1980 to 2004					

a. Complete the table by computing the percentage changes of the cost of the basket of consumer goods and the nominal wages.

b. How do the percentage changes in nominal wages compare to the percentage change in the cost of consumer goods?

c. Which sectors experienced an increase in real wages, and which sectors experienced a decrease in real wages?

5.9 **Repaying a Car Loan.** Suppose you borrow money to buy a car and must repay $20,000 in interest and principal in 5 years. Your current monthly salary is $4,000. (Related to Application 5 on page 70.)

a. Complete the following table.

b. Which environment has the lowest real cost of repaying the loan?

Change in Prices and Wages	Monthly Salary	Months of Work to Repay $20,000 Loan
Stable	$4,000	
Inflation: Prices rise by 25%		
Deflation: Prices drop by 50%		

5.10 **Inflation and Interest Rates.** Len buys MP3 music at $1 per tune and prefers music now to music later. He is willing to sacrifice 10 tunes today as long as he gets at least 11 tunes in a year. When Len loans $50 to Barb for a 1-year period, he cuts back his music purchases by 50 tunes.

a. To make Len indifferent about making the loan, Barb must repay him __________ tunes or $__________. The implied interest rate is __________ percent.

b. Suppose that over the 1-year period of the loan, all prices (including the price of MP3 tunes) increase by 20 percent, and Len and Barb anticipate the price changes. To make Len indifferent about making the loan, Barb must repay him __________ tunes or $__________. The implied interest rate is __________ percent.

ECONOMIC EXPERIMENT

PRODUCING FOLD-ITS

Here is a simple economic experiment that takes about 15 minutes to run. The instructor places a stapler and a stack of paper on a table. Students produce "fold-its" by folding a page of paper in thirds and stapling both ends of the folded page. One student is assigned to inspect each fold-it to be sure that it is produced correctly. The experiment starts with a single student, or worker, who has 1 minute to produce as many fold-its as possible. After the instructor records the number of fold-its produced, the process is repeated with two students, three students, four students, and so on. How does the number of fold-its change as the number of workers increases?

MyLab Economics

For additional economic experiments, please visit *www.myeconlab.com.*

NOTES

1. "Rent a Tree," *The Economist*, March 3, 2008.

2. United Nations Development Program, *Human Development Report 1994* (New York: Oxford University Press, 1994); Linda Bilmes and Joseph Stiglitz, "The Economic Costs of the Iraq War: An Appraisal Three Years after the Beginning of the Conflict," *Faculty Research Working Papers*, Harvard University, January 2006; Center for American Progress, "The Opportunity Costs of the Iraq War," August 25, 2004; Scott Wallsten and Katrina Kosec, "The Economic Costs of the War in Iraq," AEI-Brookings Joint Center for Regulatory Studies, September 2005; Joseph Stiglitz and Linda Bilmes, *The Three Trillion Dollar War* (New York: W.W. Norton, 2008).

3. Colin Kennedy, "Lord of the Screens," in *Economist: The World in 2003* (London, 2003), 29.

4. Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Book 1, Chapter 2.

5. Edward Castronova, *Synthetic Worlds: The Business and Culture of Online Games* (Chicago: University of Chicago Press, 2005).

CHAPTER

3 Exchange and Markets

The nation of Latvia is located on the shore of the Baltic Sea, and since the fourteenth century has been a major commercial hub between west and east.

Latvia was at the center of the Hanseatic League, the world's first Free Trade Area. But in the early 1990s, after several decades of being part of the Soviet Union, Latvia suffered from low labor productivity in the production of most goods. The following table shows output per worker for Latvia and the European Community (EC, a group of six nearby European countries).

	European Community	Latvia
Saw timber	1200	200
Milk	250	15
Livestock	100	4
Grain	200	10

Latvia's prospects for trade with the EC appear to be dim: Why would a member of the EC purchase goods produced by such low-productivity workers? Why not just use more productive EC workers to produce everything?

CHAPTER OUTLINE AND LEARNING OBJECTIVES

MyLab Economics

MyLab Economics helps you master each objective and study more efficiently.